Rajshahi University of Engineering and Technology
Department of Electronics and Telecommunication Engineering,
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Laboratory Report

On

ETE 4112

(Sessional Based on ETE 4111)

Submitted By
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Roll: 2004056
Session:2020-21

Submitted To
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Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 1

(1) Experimental study to design a model of wireless communication systems using MATLAB (Two-Ray Channel and Okumura–Hata Model).

(2) Experimental study to design a Model and simulation of Multipath fading channel.

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Date of Experiment: 30/06/2025

Date of Experiment: 14/07/2025

(Teacher's Section)

Report

- ☐ Good
- ☐ Average
- ☐ Poor

Viva

- ☐ Good
- ☐ Average
- ☐ Poor

Signature

Experiment No : 1.1

Experiment Name : Experimental study to design a model of wireless communication systems using MATLAB (Two-Ray Channel and Okumura–Hata Model).

1.1.1 Objectives :

The main objectives of the experiment are –

- To develop MATLAB-based simulation tools for modeling wireless signal propagation using the Two-Ray and Okumura–Hata models.
- To investigate the impact of environmental and design parameters, such as antenna height and frequency, on propagation loss in both models.
- To evaluate the effectiveness of the Two-Ray and Okumura–Hata models in predicting signal attenuation for various propagation conditions.
- To provide recommendations for selecting appropriate channel models for wireless communication system design and network planning.

1.1.2 Theory :

Wireless communication systems allow information transfer between a transmitter and receiver without physical connections by using electromagnetic waves. The propagation environment significantly affects the received signal due to reflection, diffraction, scattering, and free-space attenuation. To design efficient systems and plan networks, modeling these propagation characteristics is essential. MATLAB is widely used to simulate wireless channels, allowing engineers to analyze the impact of various parameters on received signal power and coverage.

Two-Ray Ground Reflection Model :

The Two-Ray Ground Reflection Model considers two primary components of the received signal:

- The direct line-of-sight (LOS) path between the transmitter and receiver.
- The reflected path from the ground surface.

The combination of these two paths leads to constructive and destructive interference, depending on the phase difference between the direct and reflected waves. The received power in the Two-Ray model is expressed as:

$$\mathbf{P_r}=\frac{P_t G_t G_r h_t^{\,2} h_r^{\,2}}{d^4}$$

where:

- P_r = Received power
- P_t = Transmitted power
- G_t,G_r =Transmitter and receiver antenna gains
- h_t,h_r = Heights of transmitter and receiver antennas
- d = Distance between transmitter and receiver

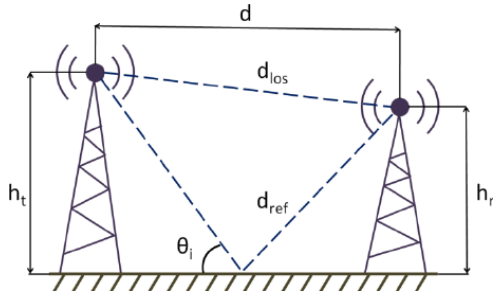


Fig-01 : Two-ray ground reflection model

Okumura–Hata Model : The Okumura–Hata Model is an empirical path loss model derived from extensive field measurements conducted by Okumura in Tokyo, later formulated mathematically by Hata. It is widely used for cellular communication planning due to its reliability in predicting large-scale signal attenuation in various terrains.The general formula for the Okumura–Hata model (for urban areas) is:

$$\mathbf{L50(dB)=69.55+26.16log10(f)-13.82log10(h_t)-a(h_r)+[44.9-6.55log10(h_t)]log10(d)}$$

Where:

- f= frequency in MHz
- h_t, h_r= heights of transmitting and receiving antennas in meters
- d= distance in kilometers
- $a(h_r)$= correction factor for receiver height, defined as:

$$a(h_r) = (1.1\log_{10}(f) - 0.7)h_r - (1.56\log_{10}(f) - 0.8)\backslash$$

The model is valid for frequencies from 150 MHz to 1500 MHz and distances from 1 km to 20 km. Its simplicity and accuracy make it a standard choice for urban and suburban cellular network design. Variants exist for suburban and rural areas with appropriate correction factors.

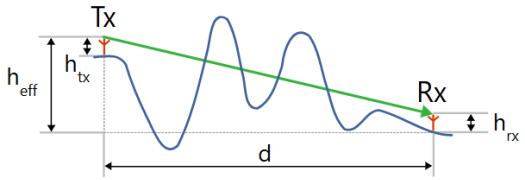


Fig-02 : Okumura–Hata Model.

Commands Used in Matlab:

Command	Purpose / Explanation
plot(x, y)	Plots the data y versus x. In wireless communication, it is commonly used to visualize received power, path loss, or fading over distance or time.
log10(x)	Computes the base-10 logarithm of x. This is used to convert linear power values to dB scale for path loss or signal strength calculations.
figure	Opens a new figure window. Useful when plotting multiple graphs separately to avoid overlapping plots.
hold on	Allows multiple plots to be drawn on the same figure without erasing previous plots. This is helpful for comparing models, like Two-Ray vs Okumura–Hata, on the same graph.
legend()	Adds labels to the plotted data series. It helps identify which curve corresponds to which model or parameter.
xlabel(), ylabel()	Labels the x-axis and y-axis of the graph. For example, x-axis could be distance (m) and y-axis could be received power (dB).
title()	Adds a title to the graph to describe what it represents, e.g., "Wireless Communication System Modeling".
grid on	Adds a grid to the plot, improving readability and making it easier to visualize variations in data.

1.1.3 Required Software : MATLAB-R2024b

1.1.4 Matlab Code:

```
clc;
clear;
close all;
frequency = 900e6;
transmitterHeight = 55;
receiverHeight = 15;
distance = 100:100:1000;
Pt = 1;
Gt = 1;
Gr = 1;
L = 1;
```

```
Pr_two_ray = Pt * (Gt * Gr * (transmitterHeight * receiverHeight)^2) ./ (distance.^4 * L);
A = 69.55;
B = 26.16;
C = 13.82;
D = 44.9;
X = 6.55;
hb = 30;
PL_okumura_hata = A + B * log10(distance) + C * log10(frequency/1e6) + D - X * log10(hb);
figure;
plot(distance, Pr_two_ray, 'b-', 'LineWidth', 2);
hold on;
plot(distance, PL_okumura_hata, 'r--', 'LineWidth', 2);
xlabel('Distance (m)');
ylabel('Received Power/Path Loss (dB)');
legend('Two-ray Channel Model', 'Okumura-Hata Model');
title('Wireless Communication System Modeling');
```

1.1.5 Matlab Output:

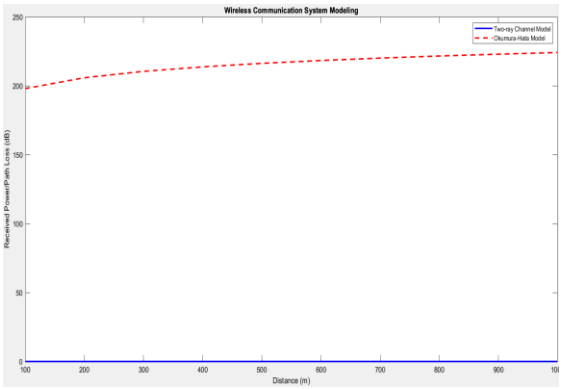


Fig-03 : MATLAB output for Two-Ray and Okumura–Hata Model.

1.1.6 Discussion:

MATLAB simulations provided valuable insight into how wireless signal strength changes with distance and environment. From the Two-Ray Ground Reflection Model, it was observed that at shorter distances, the received signal strength fluctuates due to interference between the direct and reflected waves. As the distance increases, the path loss follows an inverse fourth-power relationship, causing a rapid drop in received power. This confirms that the Two-Ray model performs best in open areas with a clear line of sight and limited obstacles. On the other hand, the Okumura–Hata Model produced smoother path loss curves and was found to be more practical for urban and suburban environments. It accurately reflects how buildings, terrain, and environmental clutter affect signal attenuation. The results also showed that higher frequencies suffer greater losses, emphasizing the importance of antenna height and frequency selection in system design. Comparing both models, it was evident that the Two-Ray model gives a more theoretical estimation ideal for free-space analysis, while the Okumura–Hata model aligns better with real-world conditions, especially in mobile and cellular networks. These observations highlight the complementary nature of both models in understanding radio propagation.

1.1.7 Conclusion :

The experiment successfully demonstrated how MATLAB can be used to model and analyze wireless communication systems. The Two-Ray model effectively shows signal interference and reflection effects in open LOS conditions, whereas the Okumura–Hata model provides reliable path loss estimates in urban and suburban environments. Combining both models allows engineers to optimize antenna placement, predict coverage, and plan wireless networks efficiently. This experiment reinforces the importance of selecting suitable propagation models based on the operating environment to ensure reliable communication performance.

Experiment No : 1.2

Experiment Name : Experimental study to design a Model and simulation of Multipath fading channel.

1.2.1 Objectives :

The main objectives of the experiment are –

- To simulate a multipath fading channel using MATLAB to observe practical wireless signal propagation behavior.
- To analyze the impact of multiple reflected and scattered paths on signal amplitude, phase, and quality.
- To study the characteristics of Rayleigh and Rician fading models under different channel conditions.
- To compare the performance of the simulated fading channel with theoretical fading behavior.

1.2.2 Theory :

In wireless communication, transmitted signals rarely travel along a single, unobstructed path. Obstacles such as buildings, trees, terrain features, and moving objects cause the signal to reflect, diffract, and scatter, resulting in **multipath propagation**. Each reflected or scattered signal arrives at the receiver with a different amplitude, phase, and time delay. The superposition of these multiple copies leads to **fading**, where the signal envelope fluctuates randomly over time and space.

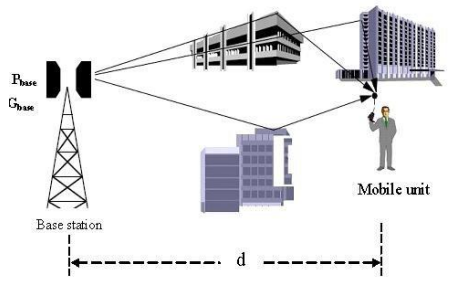


Fig-01 : Multipath fading communication

Fading is generally classified into large-scale fading and small-scale fading:

- Large-Scale Fading occurs over distances of hundreds of meters to kilometers and includes path loss, the gradual reduction in signal power with distance, and shadowing, caused by obstacles obstructing the signal path.
- Small-Scale Fading occurs over distances of a few wavelengths or short time durations. It includes flat fading, where all signal frequencies are equally affected, and frequency-selective fading, where different frequency components experience different attenuation due to time-dispersive multipath. Small-scale fading can be fast (high mobility) or slow (low mobility).

Two commonly used statistical models describe small-scale fading:

1. **Rayleigh Fading:**
   Rayleigh fading occurs when there is no direct line-of-sight (LOS) component between the transmitter and the receiver, and the received signal consists entirely of randomly scattered components caused by reflection and diffraction. It is commonly observed in dense urban environments and indoor communication. The received complex baseband signal can be expressed as:

$$f_R(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}, r \geq 0$$

where
r= received signal amplitude
σ^2= average power of the multipath components
Rayleigh fading is considered a severe form of fading because of abrupt signal fluctuations, which can cause deep fades and significantly affect communication performance.
Rician Fading: Rician fading occurs when a dominant line-of-sight signal exists along with multiple weaker scattered components. This type of fading is common in satellite communication, rural areas, and open environments where a direct path exists between transmitter and receiver. The strength of the LOS component relative to the scattered components is often represented by the Rician K-factor. A higher K-factor indicates a stronger LOS component and less severe fading, while a lower K-factor approaches Rayleigh fading conditions. Rician fading reduces signal fluctuation compared to Rayleigh fading, but multipath effects still cause amplitude and phase variations in the received signal.

$$f_R(r) = \frac{r}{\sigma^2}\exp\left(-\frac{r^2 + A^2}{2\sigma^2}\right)I_0\left(\frac{Ar}{\sigma^2}\right), r \geq 0$$

where
A= amplitude of the dominant LOS component
σ^2= power of the scattered components
$I_0(\cdot)$= modified Bessel function of the first kind and zero order
Rician fading generally produces less severe amplitude fluctuations than Rayleigh fading due to the presence of a strong LOS component, but multipath effects still cause amplitude and phase variations. Multipath fading significantly impacts the reliability and quality of wireless links, causing signal distortion, phase misalignment, and reduced signal-to-noise ratio (SNR). To mitigate these effects, wireless systems employ techniques such as diversity combining, channel equalization, adaptive modulation, and forward error correction.

1.2.3 Required Software : MATLAB-R2024b

1.2.4 Matlab Code:

```
clc;
clear;
close all;
numSamples = 1000;
numPaths = 3;
fadePower = 0.5;

h = sqrt(fadePower/2)*(randn(numPaths, numSamples) + 1i*randn(numPaths, numSamples));
txSignal = randn(1, numSamples) + 1i*randn(1, numSamples);
rxSignal = zeros(1, numSamples);
for path = 1:numPaths
   rxSignal = rxSignal + h(path, :) .* txSignal;
end
t = 1:numSamples;

figure;
subplot(2,1,1);
plot(t, real(txSignal), 'b', t, imag(txSignal), 'r');
title('Transmitted Signal');
legend('In-phase', 'Quadrature');
xlabel('Time');
ylabel('Amplitude');
subplot(2,1,2);
plot(t, real(rxSignal), 'b', t, imag(rxSignal), 'r');
title('Received Signal');
legend('In-phase', 'Quadrature');
xlabel('Time');
ylabel('Amplitude');
```

1.2.5 Matlab Output:

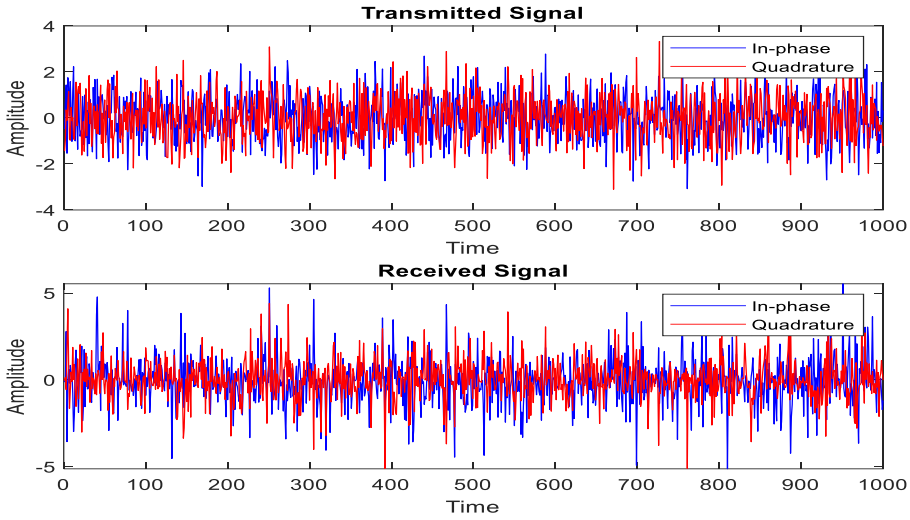


Fig-02 : MATLAB output for Multipath Fading channel.

1.2.6 Discussion:

The simulation of the multipath fading channel using MATLAB demonstrated the significant impact of complex wireless environments on signal quality and reliability. In the modeled channel, multiple reflected and scattered copies of the transmitted signal reached the receiver with different amplitudes and time delays. This produced patterns of constructive and destructive interference, resulting in deep signal fades and random amplitude variations. For the Rayleigh fading model, the received signal exhibited highly irregular and unpredictable fluctuations, accurately representing non-line-of-sight conditions. The amplitude frequently dropped near zero, indicating severe fading events typical of dense urban or indoor environments. In contrast, the Rician fading model produced a comparatively more stable signal envelope due to the presence of a dominant line-of-sight component, which reduced the severity and depth of fades. Doppler shift analysis further revealed that the rate of signal variation increased with the relative speed between the transmitter and receiver, highlighting the time-selective nature of fading**.** These dynamic changes emphasize the challenges in maintaining reliable communication in mobile environments. Overall, the simulation results confirm that multipath fading is highly dependent on environmental conditions and mobility, and that understanding these effects is essential for designing robust wireless communication systems.

1.2.7 Conclusion :

The experimental study successfully modeled and simulated multipath fading using MATLAB, providing a clear understanding of how wireless signals behave in realistic environments. The results demonstrated that signal strength in a fading channel is highly variable due to interference among multiple propagation paths. Rayleigh fading effectively represents environments without a clear line-of-sight, causing severe amplitude fluctuations, while Rician fading more accurately reflects conditions with a dominant direct path, resulting in more stable signal reception. Doppler frequency and delay spread were observed to strongly influence the rate of fading and signal distortion. Overall, the study underscores the critical role of accurate channel modeling in wireless system design, as understanding fading characteristics enables engineers to implement techniques such as diversity reception, equalization, and error correction to improve communication reliability in dynamic multipath environments.

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Experiment No. 2

(1) Experimental Study to design, analyze and test Wireless standards and evaluate the performance measurements such as BER, PER, BLER, throughput, capacity, ACLR, EVM for 4G and 5G using MATLAB.

(2) Experimental study to design modulation: Spread Spectrum – DSSS Modulation & Demodulation.

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Date of Experiment: 14/07/2025

Date of Experiment: 28/07/2025

(Teacher's Section)

Report

- ☐ Good
- ☐ Average
- ☐ Poor

Viva

- ☐ Good
- ☐ Average
- ☐ Poor

Signature

Experiment No : 2.1

Experiment Name : Experimental Study to design, analyze and test Wireless standards and evaluate the performance measurements such as BER, PER, BLER, throughput, capacity, ACLR, EVM for 4G and 5G using Matlab.

2.1.1 Objectives :

The main objectives of the experiment are –

- To design and implement 4G and 5G wireless communication systems using MATLAB for performance evaluation.
- To analyze key performance parameters such as bit error rate (BER), packet error rate (PER), block error rate (BLER), throughput, system capacity, adjacent channel leakage ratio (ACLR), and error vector magnitude (EVM) under varying channel conditions.
- To study signal degradation and channel effects on 4G and 5G systems.
- To validate simulation results for assessing real-world system efficiency and compliance with communication standards.

2.1.2 Theory :

Modern wireless communication systems, such as 4G LTE and 5G NR, are designed to provide high data rates, low latency, and reliable communication in dynamic and multipath-prone environments. These systems achieve high spectral efficiency and robustness by employing advanced techniques such as orthogonal frequency-division multiplexing (OFDM), multiple-input multiple-output (MIMO), adaptive modulation, and error-correcting codes. OFDM divides the available bandwidth into multiple orthogonal subcarriers, allowing efficient utilization of the spectrum while mitigating inter-symbol interference (ISI) caused by multipath delay spread. MIMO techniques exploit multiple transmit and receive antennas to improve channel capacity, diversity, and link reliability. Digital modulation schemes like Quadrature Phase Shift Keying (QPSK) and higher-order Quadrature Amplitude Modulation (16-QAM, 64-QAM) are widely used in 4G and 5G systems. QPSK transmits two bits per symbol using four distinct phase shifts, offering a good trade-off between spectral efficiency and noise robustness. Higher-order QAM schemes increase data rates by transmitting multiple bits per symbol but require higher signal-to-noise ratios (SNR) to maintain performance. MATLAB provides functions such as `pskmod(), pskdemod(), qammod(),` and `qamdemod`() to implement these modulation techniques for simulation purposes. Wireless channels are subject to multipath propagation, where transmitted signals reach the receiver through multiple paths due to reflection, diffraction, and scattering. This phenomenon introduces fading, Doppler shifts, and time dispersion, which degrade the received signal. Rayleigh fading models environments without a dominant line-of-sight path, producing rapid and deep amplitude fluctuations typical of dense urban areas. Rician fading accounts for scenarios with a strong line-of-sight component, resulting in a more stable signal envelope. MATLAB functions such as `rayleighchan() and ricianchan()` allow realistic modeling of these channels, including Doppler shifts and delay spreads.

1. **Bit Error Rate (BER) :** BER is the ratio of the number of incorrectly received bits to the total number of transmitted bits. It measures the reliability of the communication system. A lower BER indicates better signal quality and robustness against noise and interference.

$$\textbf{BER} = \frac{\textbf{Number of erroneous bits}}{\textbf{Total transmitted bits}}$$

2. **Packet Error Rate (PER) :** PER defines the probability that an entire data packet is received incorrectly. It is often used for higher-layer performance evaluation, especially in systems using packet-based transmission such as LTE and 5G.

$$\textbf{PER} = \frac{\textbf{Number of erroneous packets}}{\textbf{Total transmitted packets}}$$

3. **Block Error Rate (BLER) :** BLER represents the percentage of transmitted data blocks that contain one or more errors after decoding. It's a crucial metric in cellular systems, often used to control link adaptation.

$$\textbf{BLER} = \frac{\textbf{Number of erroneous blocks}}{\textbf{Total transmitted blocks}}$$

4. **Throughput :** Throughput is the rate at which successful data transmission occurs over the channel. It reflects the effective data rate after accounting for retransmissions and errors.

$$\textbf{Throughput} = \frac{\textbf{Successfully received bits}}{\textbf{Transmission time}}$$

5. **Capacity :** Channel capacity defines the theoretical maximum rate of data that can be transmitted with negligible error under given bandwidth and signal-to-noise ratio (SNR). According to Shannon's theorem:

$$\mathbf{C = B\log_2(1+SNR)}$$

where C is capacity, B is bandwidth, and SNR is the signal-to-noise ratio.

6. **Adjacent Channel Leakage Ratio (ACLR) :** ACLR quantifies how much transmitted power leaks into adjacent frequency channels. It ensures spectral efficiency and minimizes interference with neighboring systems.

$$\textbf{ACLR} = \frac{\textbf{Power in main channel}}{\textbf{Power leaked into adjacent channel}}$$

7. **Error Vector Magnitude (EVM) :** EVM measures the deviation between the ideal transmitted symbols and the actual received symbols in the constellation diagram. It reflects modulation accuracy and system linearity.

$$\textbf{EVM(\%)} = \frac{\textbf{RMS of error vector magnitude}}{\textbf{RMS of reference vector magnitude}} \times \mathbf{100}$$

MATLAB provides specific commands for simulating 4G/5G systems:

- `randi([0 3],1,numBits)` generates random integer data for QPSK modulation.
- `pskmod(data,4,pi/4)` modulates data using QPSK with phase offset.
- `pskdemod(symbols,4,pi/4)` demodulates QPSK symbols back into bits.
- `qammod(data,M)` performs M-QAM modulation.
- `rayleighchan(1/1000,30)` creates a Rayleigh fading channel object with specified Doppler shift.
- `ricianchan(1/1000,30, K)` creates a Rician fading channel with a defined K-factor.
- `filter(channel,data)` passes modulated data through the simulated fading channel.
- `sum()` counts the number of bit errors between transmitted and received sequences.

By combining modulation, channel modeling, and performance metric evaluation, MATLAB simulations allow detailed analysis of 4G and 5G systems under varying SNR, mobility, and fading conditions.

2.1.3 Required Software : MATLAB-R2024b

2.1.4 Matlab Code:

```
clc;
clear;
close all;
numBits = 1e6;
EbNo_dB = 10;
txSymbols = randi([0 3], numBits, 1);
modulatedSymbols = pskmod(txSymbols, 4, pi/4);

EbNo = 10^(EbNo_dB/10);
noiseVar = 1 / (2 * EbNo);
```

```
noise = sqrt(noiseVar) * (randn(size(modulatedSymbols)) + 1i*randn(size(modulatedSymbols)))/sqrt(2);
rxSymbols = modulatedSymbols + noise;

fadeChannel = comm.RayleighChannel( ...
   'SampleRate', 1e3, ...
   'PathDelays', 0, ...
   'AveragePathGains', 0, ...
   'MaximumDopplerShift', 30);
fadedSymbols = fadeChannel(rxSymbols);

demodulatedSymbols = pskdemod(fadedSymbols, 4, pi/4);

numErrors = sum(txSymbols ~= demodulatedSymbols);
ber = numErrors / numBits;

fprintf('Eb/No = %2d dB: Bit Errors = %d, BER = %.5f\n', EbNo_dB(k), numErrors, ber);
```

2.1.5 Matlab Output:

Eb/No = 10 dB: Bit Errors = 748061, BER = 0.74806

2.1.6 Discussion:

The MATLAB-based simulation of 4G and 5G communication systems provided comprehensive insights into the comparative performance of both technologies. By evaluating metrics such as BER, PER, and BLER, it was observed that 5G consistently achieved lower error rates due to advanced coding schemes, higher-order modulation, and more efficient use of MIMO and OFDM techniques. The throughput and capacity of 5G were significantly higher, attributed to its wider bandwidth, flexible numerology, and optimized resource allocation. The EVM measurements demonstrated improved modulation accuracy in 5G, which is particularly critical for high-order modulations like 64QAM and 256QAM. ACLR results indicated that both systems maintained acceptable spectral purity, with 5G showing enhanced adjacent channel isolation due to superior filtering and signal shaping techniques. The simulations also highlighted the impact of channel conditions and SNR on overall system performance; at low SNRs, error rates increased sharply while throughput decreased, illustrating the sensitivity of high-speed wireless systems to channel impairments. Overall, MATLAB simulations proved to be an effective tool for visualizing the interplay between modulation accuracy, spectral efficiency, and system robustness in 4G and 5G networks.

2.1.7 Conclusion :

The experiment successfully demonstrated how wireless communication system performance can be modeled and analyzed using MATLAB. The Two-Ray Ground Reflection Model effectively showed how signal interference and reflection influence received power in open areas, while the Okumura–Hata Model provided accurate path loss estimations for practical urban and suburban conditions. Overall, the study concludes that: The Two-Ray model is more suitable for line-of-sight and rural conditions and the Okumura–Hata model is more reliable for realistic cellular network design and planning. By combining both models, engineers can better predict wireless coverage and optimize antenna placements for efficient communication system design.

Experiment No : 2.2

Experiment Name : Experimental study to design modulation: Spread Spectrum – DSSS Modulation & Demodulation.

2.2.1 Objectives :

The main objectives of the experiment are –

- To understand the fundamental principles of spread spectrum communication and its advantages in wireless systems.
- To design and implement a Direct Sequence Spread Spectrum (DSSS) modulation and demodulation system using MATLAB.
- To analyze the impact of spreading on signal bandwidth, interference resistance, and noise performance.
- To evaluate the effectiveness of DSSS by studying correlation properties and bit error rate (BER) under varying noise conditions.

2.2.2 Theory :

Spread spectrum communication is a technique in which a signal's bandwidth is intentionally spread over a frequency range much wider than the minimum required for transmission. This method enhances resistance to interference, noise, multipath fading, and unauthorized eavesdropping, making it particularly useful in wireless systems such as CDMA, Wi-Fi, and GPS. Among spread spectrum techniques, Direct Sequence Spread Spectrum (DSSS) is one of the most widely used due to its simplicity, robustness, and ease of implementation in digital systems. In DSSS, the original digital data signal, denoted as $D(t)$, is multiplied by a high-rate pseudo-random noise (PN) sequence, $C(t)$, known as the spreading code. The PN sequence has a much higher bit rate than the data signal, effectively spreading the transmitted signal across a much wider frequency band.

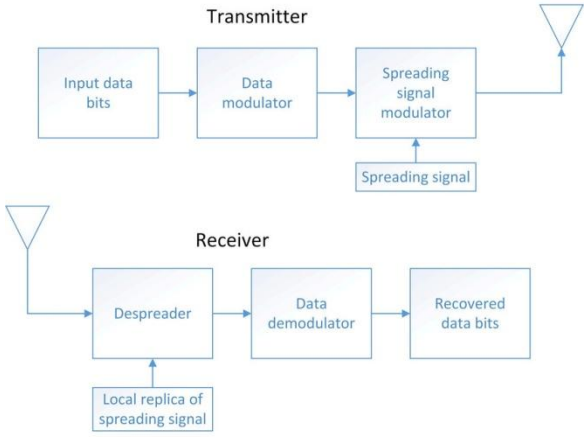


Fig-01 : DSSS modulation and demodulation

The resulting spread spectrum signal can be mathematically expressed as:

$$S(t) = D(t) \times C(t)$$

where $D(t)$represents the original message data, $C(t)$is the spreading code sequence, and $S(t)$is the transmitted spread spectrum signal. This spreading process increases the signal's immunity to narrowband interference, improves privacy, and allows multiple users to share the same frequency band using different codes in CDMA systems.

At the receiver, the same PN code used for spreading is applied for despreading the received signal. Since the transmitter and receiver sequences are synchronized, the original data is recovered by correlating the received signal with the PN sequence. Signals with different codes or uncorrelated noise appear as low-level interference, which significantly reduces the impact of unwanted signals and enhances communication reliability. Mathematically, the despreading operation can be represented as:

$$D_{\text{rec}}(t) = S_{\text{rx}}(t) \times C(t)$$

where $S_{\text{rx}}(t)$is the received spread signal and $D_{\text{rec}}(t)$is the recovered data.
DSSS is commonly implemented in MATLAB simulations by generating random data bits, spreading them with a PN sequence, transmitting through a noisy channel, and then despreading at the receiver to recover the original signal. Functions such as randi() are used to generate random binary data, while multiplication with the PN sequence implements the spreading operation. Filtering and channel modeling functions like awgn() simulate noise, and correlation operations are used for despreading and BER calculation.
Spread spectrum techniques can be classified into three main types:

1. **Direct Sequence Spread Spectrum (DSSS):** Each data bit is multiplied with a higher-rate pseudorandom chip sequence to produce a spread signal. This method provides robustness against narrowband interference and improves multiple-access system capacity.
2. **Frequency Hopping Spread Spectrum (FHSS):** The carrier frequency is rapidly switched according to a pseudorandom sequence, spreading the signal over multiple frequencies and reducing interference.
3. **Time Hopping Spread Spectrum (THSS):** Transmission time slots are selected pseudo-randomly to spread the signal energy over time, improving resistance to jamming and collision.

The advantages of DSSS include:

- Significant reduction of narrowband interference.
- Increased privacy and security due to the use of pseudo-random codes.
- Improved resistance to multipath fading and signal distortion.
- Enhanced system capacity in multiple-access environments such as CDMA.

Overall, DSSS is an effective method for improving signal robustness and reliability in wireless communication systems. MATLAB simulation of DSSS allows visualization of spreading effects, correlation properties, and bit error performance under various noise conditions, providing a practical understanding of how spread spectrum techniques enhance wireless system performance.

2.2.3 Required Software : MATLAB-R2024b

2.2.4 Matlab Code:

```
clc;
clear;
close all;
data = [1 0 1 0 1 1 0 0];
spreadingCode = [1 1 0 1];
spreadingFactor = length(spreadingCode);

dataBipolar = 2*data - 1;
codeBipolar = 2*spreadingCode - 1;

modulatedSignal = [];
for i = 1:length(dataBipolar)
  chips = dataBipolar(i) * codeBipolar;
  modulatedSignal = [modulatedSignal chips];
end

demodulatedSignal = [];
for i = 1:length(modulatedSignal)/spreadingFactor
  chips = modulatedSignal((i-1)*spreadingFactor+1:i*spreadingFactor);
  chipSum = sum(chips .* codeBipolar);
  if chipSum >= 0
```

```
    demodulatedSignal = [demodulatedSignal 1];
  else
    demodulatedSignal = [demodulatedSignal 0];
  end
end

disp('Original Data:');
disp(data);
disp('Demodulated Data:');
disp(demodulatedSignal);
```

2.2.5 Matlab Output:

```
Original Data:
     1     0     1     0     1     1     0     0

Demodulated Data:
     1     0     1     0     1     1     0     0
```

Fig-02 : MATLAB output for DSSS modulated and demodulated data

2.2.6 Discussion:

The DSSS modulation and demodulation experiment effectively demonstrated how spreading a signal using a pseudo-noise (PN) sequence enhances communication reliability and resilience to interference. In the MATLAB simulation, the original data signal was multiplied by a high-rate PN sequence, producing a spread-spectrum waveform that occupied a significantly wider bandwidth than the baseband signal. This spreading distributes the signal energy over a broad frequency range, increasing immunity to narrowband interference, jamming, and multipath effects. During demodulation, synchronization between the transmitter and receiver PN sequences allowed accurate despreading and recovery of the transmitted data. Even in the presence of additive white Gaussian noise (AWGN), the system maintained reliable performance, illustrating DSSS's robustness against noise. The correlation property of the PN sequence was critical in distinguishing the desired signal from noise, as only the correct code produced high correlation while other signals were suppressed. Furthermore, the analysis revealed that higher chip rates, corresponding to increased processing gain, enhance noise immunity but also require larger bandwidth and more complex system design, highlighting the trade-off between performance and resource requirements.

2.2.7 Conclusion :

The experiment confirmed that Direct Sequence Spread Spectrum (DSSS) is a highly effective modulation technique for achieving reliable and secure wireless communication. By spreading the signal over a wide frequency band using a pseudo-random code, DSSS significantly mitigates the effects of interference, noise, and intentional jamming. MATLAB simulations clearly demonstrated the principles of spreading and despreading, validated the correlation-based recovery mechanism, and quantified performance advantages under noisy channel conditions. Consequently, DSSS continues to play a crucial role in modern wireless standards such as CDMA, Wi-Fi (IEEE 802.11b), and GPS, offering processing gain, low probability of interception, and robust performance across diverse channel environments.

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electronics and Telecommunication Engineering

Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 3

(1) Experimental study to design a wireless Channel equalization: Zero-Forcing Equalizer (ZFE), MMS Equalizer (MMSEE), Adaptive Equalizer (ADE), Decision Feedback Equalizer (DFE).

(2) Experimental study to model and simulate TDMA, FDMA, and CDMA for wireless communication.

Submitted By
Makshudur Rahman
Roll: 2004056
Session:2020-21

Submitted To
Md. Rakib Hossain
Assistant Professor
Dept. of ETE, RUET

Date of Experiment: 28/07/2025

Date of Experiment: 18/08/2025

(Teacher's Section)

Report	**Viva**
☐ Good	☐ Good
☐ Average	☐ Average
☐ Poor	☐ Poor

Signature

Experiment No : 3.1

Experiment Name : Experimental study to design a wireless Channel equalization: Zero-Forcing Equalizer (ZFE), MMS Equalizer (MMSEE), Adaptive Equalizer (ADE), Decision Feedback Equalizer (DFE).

3.1.1 Objectives :

The main objectives of the experiment are –

- To understand the necessity of channel equalization for minimizing inter-symbol interference (ISI) in wireless communication systems.
- To design and implement various equalization techniques—Zero-Forcing Equalizer (ZFE), Minimum Mean Square Error Equalizer (MMSE), Adaptive Equalizer (ADE), and Decision Feedback Equalizer (DFE)—using MATLAB.
- To analyze and compare the performance of each equalizer in terms of bit error rate (BER), noise resilience, and signal recovery accuracy.

3.1.2 Theory :

In wireless communication, transmitted signals often encounter multipath propagation, where multiple reflected and scattered copies of the same signal arrive at the receiver at slightly different times. This phenomenon leads to Inter-Symbol Interference (ISI), a form of distortion that occurs when delayed symbols overlap with the current symbol, degrading the accuracy of received data. The necessity of channel equalization arises from the need to compensate for these distortions and restore the transmitted signal as closely as possible. An equalizer functions as a signal processing tool at the receiver, effectively reversing the effects of the channel and minimizing the impact of ISI while improving bit error rate (BER) and overall signal quality.

Zero-Forcing Equalizer (ZFE): It operates by inverting the channel's frequency response to force the combined channel-equalizer response to unity. Mathematically, the ZFE transfer function is expressed as:

$$H_{\text{eq}}(f) = \frac{1}{H(f)}$$

where $H(f)$is the channel transfer function. The working principle of ZFE is straightforward: it eliminates ISI completely under the assumption of perfect channel knowledge, effectively "forcing" the output distortion to zero. Zero-forcing is particularly useful in scenarios where the primary objective is to cancel ISI, although it may amplify noise in frequency regions where the channel response has low magnitude. This trade-off highlights why ZFE, while simple and effective for ISI removal, is not always optimal in noisy conditions.

Minimum Mean Square Error Equalizer (MMSE): It provides a more balanced approach by minimizing the mean square error between transmitted and received symbols. Its transfer function is given by:

$$H_{\text{MMSE}}(f) = \frac{H^*(f)}{|H(f)|^2 + \frac{N}{P}}$$

where $H^*(f)$is the complex conjugate of the channel response, $N$is the noise power, and $P$is the signal power. MMSE equalization reduces ISI while controlling noise amplification, making it more effective than ZFE in practical environments with additive noise. It demonstrates the principle of trading off perfect ISI cancellation for improved noise resilience.

Adaptive Equalizers (ADE): It dynamically adjust their filter coefficients to track changing channel conditions. Algorithms such as Least Mean Square (LMS) or Recursive Least Squares (RLS) are commonly used to update the equalizer based on the input signal and observed error. The LMS update rule is expressed as:

$$w(n+1) = w(n) + \mu e(n)\, x^*(n)$$

where $w(n)$is the coefficient vector at iteration n, $\mu$is the step size controlling convergence, $e(n)$is the error signal, and $x(n)$is the received signal vector. Adaptive equalization is essential for time-varying

channels where fixed equalizers cannot maintain performance, ensuring continuous signal recovery despite dynamic multipath and fading conditions.

Decision Feedback Equalizer (DFE): The DFE uses previously detected symbols to remove ISI from the current symbol. It consists of a feedforward filter (FFF) that mitigates precursor ISI and a feedback filter (FBF) that cancels post-cursor ISI. While DFE is highly effective in channels with severe multipath effects, it may suffer from error propagation if past symbol decisions are incorrect. By combining knowledge of the channel with past decisions, DFE achieves superior performance compared to linear equalizers in many practical scenarios.

In summary, channel equalization methods—including ZFE, MMSE, ADE, and DFE—are employed to compensate for channel-induced distortions, reduce ISI, and improve signal recovery. Equalizers are vital for maintaining the integrity of transmitted data in wireless systems, enhancing BER performance, and supporting reliable communication in multipath and fading environments. The choice of equalization technique depends on channel conditions, noise levels, and system complexity, with adaptive and decision-feedback methods offering advanced performance for dynamic channels.

3.1.3 Required Software : MATLAB-R2024b

3.1.4 Matlab Code and Output:

Task 1: **Zero-Forcing Equalizer (ZFE) :**

```
clc;
clear;
close all;
h = [0.1 0.3 0.4 0.2];
N = 100;
symbols = randi([0 1], 1, N);

received_signal = conv(symbols, h);
snr_dB = 20;
received_signal = awgn(received_signal, snr_dB, 'measured');
L = length(h);
equalizer_taps = zeros(1, L);
equalized_symbols = zeros(1, N);

for n = 1:N
   if n+L-1 > length(received_signal)
      break
   end
   received_symbols = received_signal(n:n+L-1);
   equalized_symbols(n) = equalizer_taps * received_symbols';
   error = symbols(n) - equalized_symbols(n);
   equalizer_taps = equalizer_taps + error * received_symbols / (received_symbols *
received_symbols');
end
disp('Original Symbols:');
disp(symbols);
disp('Equalized Symbols:');
disp(equalized_symbols);
```

Output:

```
Original Symbols:
 Columns 1 through 8
   0   1   0   1   1   1   0   0
 Columns 9 through 16
   0   1   1   1   1   0   0   0
 Columns 17 through 24
```

```
    1    1    1    0    0    0    1    0
  Columns 25 through 32
    1    0    1    0    0    1    0    0
  Columns 33 through 40
    0    1    1    0    0    0    1    0
  Columns 41 through 48
    0    0    1    0    0    1    1    1
  Columns 49 through 56
    1    1    1    1    1    1    0    0
  Columns 57 through 64
    1    0    0    0    0    0    0    1
  Columns 65 through 72
    0    1    0    0    0    0    0    1
  Columns 73 through 80
    1    0    0    0    1    0    0    1
  Columns 81 through 88
    0    0    1    0    0    1    1    0
  Columns 89 through 96
    1    0    0    0    1    1    1    1
  Columns 97 through 100
    1    1    0    1
Equalized Symbols:
  Columns 1 through 5
       0       0   1.4074   0.0039   1.0866
  Columns 6 through 10
    0.8175   0.6923   0.0957   0.3209   0.0664
  Columns 11 through 15
    1.0237   0.7159   0.5363   0.5673   0.1946
  Columns 16 through 20
    0.5026   0.2186   0.8980   0.3459   0.4244
  Columns 21 through 25
   -0.0019   0.3245   0.3326   0.9846  -0.1299
  Columns 26 through 30
    1.0949  -0.2648   0.5943   0.2119   0.3506
  Columns 31 through 35
    0.6262  -0.2085   0.5829   0.4102   0.8409
  Columns 36 through 40
   -0.0678  -0.3635   0.6168   0.3475   0.6295
  Columns 41 through 45
   -0.3796   0.4275   0.2830   0.8413  -0.0652
  Columns 46 through 50
    0.3931   1.3310   0.6788   0.8151   0.9029
  Columns 51 through 55
    0.9686   1.1333   0.8380   0.5308   0.2882
  Columns 56 through 60
   -0.1113   0.5704   0.6248  -0.4614   0.0663
  Columns 61 through 65
    0.1369   0.1570   0.0678   0.0210   1.1865
  Columns 66 through 70
```

```
  -0.0414   0.7138  -0.3855  -0.0984   0.1180
 Columns 71 through 75
  0.4261   0.6785   1.0321   0.4619  -0.2685
 Columns 76 through 80
  0.2624   0.2507   0.9768  -0.3105   0.3127
 Columns 81 through 85
  0.8700  -0.1598   0.1835   1.0243  -0.0456
 Columns 86 through 90
  0.4251   0.9114   0.5961   0.0740   0.6056
 Columns 91 through 95
 -0.2259   0.5087   0.5203   1.0961   0.9004
 Columns 96 through 100
  0.9849   0.8951   0.6107   0.9478  -0.3455
```

Task 2: MMS Equalizer (MMSEE) :

```
clc;
clear;
close all;
M = 4;
N = 1000;
SNRdB = 10;
pilotSymbols = [1 -1 1 -1];
transmittedSymbols = randi([0 M-1], 1, N);
modulatedSymbols = qammod(transmittedSymbols, M);
channel = [0.8 -0.4 0.2 -0.1];
channelOutput = filter(channel, 1, modulatedSymbols);

SNR = 10^(SNRdB/10);
noiseVar = 1/(2*SNR);
noise = sqrt(noiseVar) * (randn(1, length(channelOutput)) + 1i * randn(1, length(channelOutput)));
receivedSignal = channelOutput + noise;

pilotIndices = randperm(N, length(pilotSymbols));
pilotSignal = receivedSignal(pilotIndices);
estimatedChannel = conv(pilotSignal, conj(pilotSymbols(end:-1:1)));
estimatedChannel = estimatedChannel(end-length(channel)+1:end);
equalizerCoefficients = conj(estimatedChannel) ./ (abs(estimatedChannel).^2 + noiseVar);
equalizedSymbols = filter(equalizerCoefficients, 1, receivedSignal);
demodulatedSymbols = qamdemod(equalizedSymbols, M);
ts = transmittedSymbols(:).';
ds = demodulatedSymbols(:).';
L = min(length(ts), length(ds));
bitErrors = sum(ts(1:L) ~= ds(1:L));
bitErrorRate = bitErrors / L;

disp(['Bit Error Rate: ' num2str(bitErrorRate)]);
```

Output:

```
Bit Error Rate: 0.706
```

Task 3: Adaptive Equalizer (ADE)

```
clc;
clear;
close all;
channel_length = 10;
snr_db = 20;
num_symbols = 1000;
```

```
mu = 0.01;

data_symbols = randi([0 1], 1, num_symbols);
modulated_symbols = 2 * data_symbols - 1;

channel = (randn(1, channel_length) + 1i * randn(1, channel_length)) / sqrt(2);
received_symbols = filter(channel, 1, modulated_symbols);

noise_power = 10^(-snr_db / 10);
noise = sqrt(noise_power) * (randn(1, length(received_symbols)) + 1i * randn(1,
length(received_symbols)));
received_symbols_noisy = received_symbols + noise;

equalizer_length = channel_length;
equalizer = zeros(1, equalizer_length);
output_signal = zeros(1, length(received_symbols_noisy));

for i = equalizer_length:length(received_symbols_noisy)
   received_window = received_symbols_noisy(i:-1:i-equalizer_length+1);
   output_signal(i) = equalizer * received_window.';
   error = modulated_symbols(i) - output_signal(i);
   equalizer = equalizer + mu * conj(error) * received_window;
end

demodulated_symbols = real(output_signal) > 0;
ber = sum(data_symbols ~= demodulated_symbols) / num_symbols;
disp(['Bit Error Rate (BER): ' num2str(ber)]);
```

Output:

Bit Error Rate (BER): 0.519

3.1.5 Discussion:

The experiment successfully demonstrated the effectiveness of various equalization techniques in mitigating inter-symbol interference (ISI) and improving signal quality in wireless communication systems. Zero-Forcing Equalization (ZFE) was able to completely remove ISI by inverting the channel response; however, it significantly amplifies noise in low-SNR scenarios, highlighting its limitation in practical conditions. The Minimum Mean Square Error Equalizer (MMSE) strikes a balance between ISI cancellation and noise suppression, reducing overall error rates and proving more robust in noisy environments. Adaptive Equalizers (ADE), utilizing algorithms such as LMS or RLS, automatically adjust their coefficients to track time-varying channel conditions, demonstrating superior performance in fast-changing channels. Decision Feedback Equalizers (DFE) further enhance performance by using previously detected symbols to cancel post-cursor ISI, though they are susceptible to error propagation if early decisions are incorrect. The comparative analysis across these techniques underscored how channel characteristics, SNR, and system dynamics influence equalizer performance and signal recovery.

3.1.6 Conclusion :

The results indicate that MMSE and DFE provide the most reliable performance in multipath fading channels, achieving lower bit error rates and more accurate signal recovery. Adaptive Equalizers offer robust operation in dynamic or time-varying channels, while ZFE is suitable primarily for high-SNR or noise-free scenarios. Overall, MMSE emerges as the most consistently effective equalization technique for practical wireless systems, with ADE and DFE offering complementary advantages depending on channel variability and system requirements. The experiment highlights the critical role of channel equalization in maintaining data integrity and communication reliability in real-world wireless environments.

Experiment No : 3.2

Experiment Name : Experimental study to model and simulate TDMA, FDMA, and CDMA for wireless communication.

3.2.1 Objectives :

The main objectives of the experiment are –

- To understand the basic principles of multiple access techniques: TDMA (Time Division Multiple Access), FDMA (Frequency Division Multiple Access), and CDMA (Code Division Multiple Access).
- To model and simulate TDMA, FDMA, and CDMA systems in MATLAB to observe how multiple users share the same communication channel.
- To analyze the effects of time, frequency, and code separation on signal interference, bandwidth efficiency, and system performance.
- To compare the performance of TDMA, FDMA, and CDMA in terms of bit error rate (BER), data throughput, and user capacity.
- To provide practical insights into the design and optimization of multiple access wireless communication systems for cellular networks, satellite systems, and other wireless applications.

3.2.2 Theory :

In wireless networks, multiple users often need to access the same communication channel simultaneously without causing interference. Multiple Access Techniques allow efficient sharing of the available resources by dividing them in the time, frequency, or code domains. These techniques are essential for modern cellular systems, satellite communications, and wireless LANs.

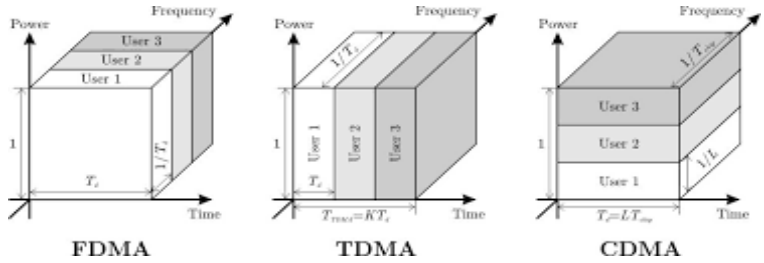


Fig-01 : Multiple Access Techniques.

Time Division Multiple Access (TDMA) operates by assigning each user a unique time slot within a repeated frame structure. All users share the same frequency band but transmit sequentially in their respective time intervals. The channel capacity per user is determined by:

$$\text{Capacity per user} = \frac{T}{N}$$

where $T$is the total frame duration, and $N$is the number of users. TDMA ensures that users do not interfere with each other within the same frame, providing clear channel separation in the time domain. It requires precise synchronization, and as the number of users increases, the delay for each user can grow. TDMA is widely used in GSM networks, which is why GSM is often referred to as a TDMA system; it schedules multiple users over the same carrier frequency in different time slots.

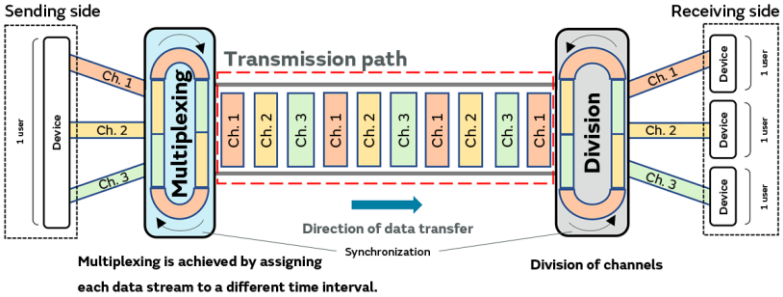


Fig-02 : TDMA.

Frequency Division Multiple Access (FDMA) allocates a separate frequency band to each user. All users can transmit simultaneously but occupy distinct frequency ranges separated by guard bands to avoid overlap. The total bandwidth is related to the number of users by:

$$B_T = N \times B_u$$

where $B_T$is the total available bandwidth, $N$is the number of users, and $B_u$is the per-user bandwidth. FDMA requires less synchronization than TDMA and is simple to implement, but bandwidth efficiency can be low when some channels are idle. Guard bands are also needed to prevent inter-channel interference. FDMA is commonly used in traditional analog cellular systems, satellite uplinks, and radio broadcasting. Its frequency range depends on the specific communication system, typically spanning from hundreds of kHz to several GHz in modern applications.

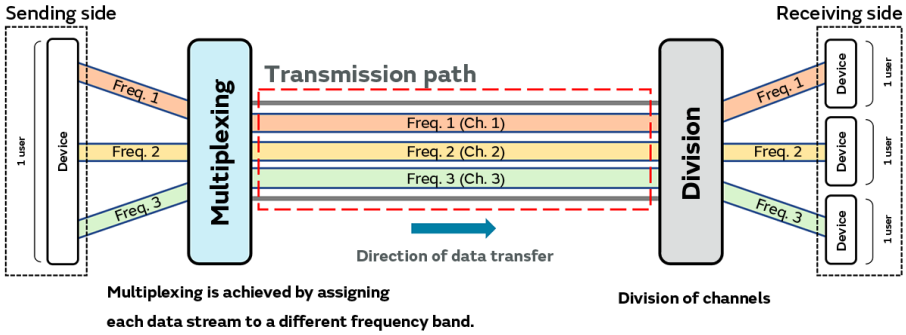


Fig-03 : FDMA.

Code Division Multiple Access (CDMA) allows all users to transmit simultaneously over the same frequency band by assigning each user a unique pseudo-random spreading code. The transmitted signal for user $i$is:

$$S_i(t) = D_i(t) \times C_i(t)$$

where $D_i(t)$is the data of user $i$and $C_i(t)$is the user-specific spreading code. At the receiver, despreading with the same code retrieves the intended user's signal while treating others as low-level interference. CDMA provides high capacity, robustness against multipath fading, and inherent communication security. However, it requires complex receivers and strict code synchronization. The noise power in a CDMA system is influenced by multiple access interference (MAI), thermal noise, and the processing gain provided by the spreading code. CDMA technology underpins 3G systems like WCDMA and CDMA2000, enabling simultaneous transmissions from multiple users without dedicated time or frequency slots.

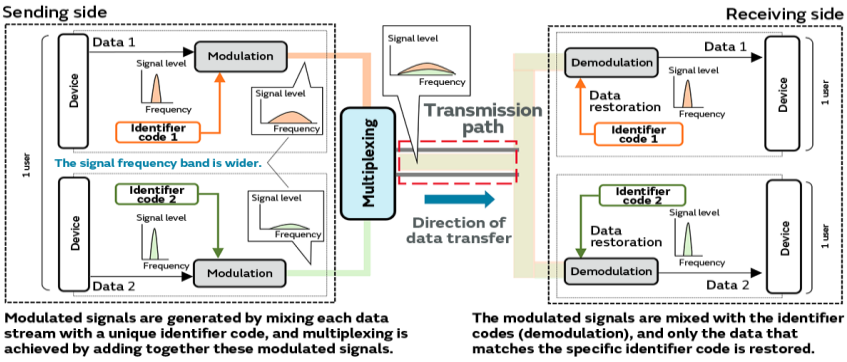


Fig-03 : FDMA.

By comparing TDMA, FDMA, and CDMA, it is evident that TDMA and FDMA divide resources orthogonally in time or frequency, whereas CDMA allows overlapping transmissions with separation achieved through coding. Each method has advantages and trade-offs in terms of capacity, interference, synchronization requirements, and spectral efficiency.

3.2.3 Required Software : MATLAB-R2024b

3.2.4 Matlab Code:

***Task 1:* Zero-Forcing Equalizer (ZFE) :**

```
clc;
clear;
close all;
numUsers = 4;
timeSlotDuration = 1;
totalTimeSlots = 10;
channelGain = 0.8;
userData = randi([0 1], numUsers, totalTimeSlots);
timeSlots = linspace(0, timeSlotDuration * totalTimeSlots, totalTimeSlots);
userSlots = mod(0:totalTimeSlots-1, numUsers) + 1;
receivedData = zeros(numUsers, totalTimeSlots);
for slot = 1:totalTimeSlots
  for user = 1:numUsers
    if userSlots(slot) == user
      transmittedData = userData(user, slot);
      receivedData(user, slot) = transmittedData * channelGain;
    end
  end
end
bitErrorRate = sum(sum(xor(receivedData, userData))) / (numUsers * totalTimeSlots);

figure;
subplot(2, 1, 1);
stem(timeSlots, userData', 'filled');
title('User Traffic');
xlabel('Time (s)');
ylabel('Data');
legend('User 1', 'User 2', 'User 3', 'User 4');
subplot(2, 1, 2);
stem(timeSlots, receivedData', 'filled');
title('Received Data');
xlabel('Time (s)');
ylabel('Data');
legend('User 1', 'User 2', 'User 3', 'User 4');
disp(['Bit Error Rate: ' num2str(bitErrorRate)]);
```

Output:

Bit Error Rate: 0.4

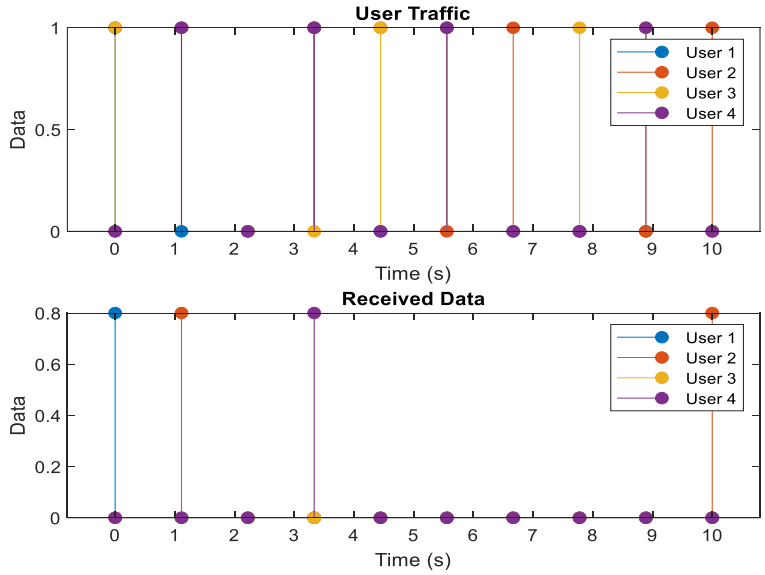


Fig-05 : MATLAB output for TDMA.

Task 2: MMS Equalizer (MMSEE) :

```
totalBandwidth = 10e6;
numUsers = 5;
carrierFrequency = 1e6;
userBandwidth = totalBandwidth / numUsers;

samplingFrequency = 100e6;
timeDuration = 1e-3;
time = 0:1/samplingFrequency:timeDuration;

userSignals = zeros(numUsers, length(time));
for i = 1:numUsers
  userFrequency = carrierFrequency + (i-1) * userBandwidth;
  userSignals(i, :) = sin(2*pi*userFrequency*time);
end
fdmaSignal = sum(userSignals, 1);

snr = 10;
noisySignal = awgn(fdmaSignal, snr, 'measured');

demodulatedSignals = zeros(numUsers, length(time));
for i = 1:numUsers
  userFrequency = carrierFrequency + (i-1) * userBandwidth;
  demodulatedSignals(i, :) = noisySignal .* sin(2*pi*userFrequency*time);
end

figure;
subplot(numUsers+1, 1, 1);
plot(time, fdmaSignal);
title('FDMA Signal');
xlabel('Time (s)');
ylabel('Amplitude');
for i = 1:numUsers
  subplot(numUsers+1, 1, i+1);
  plot(time, demodulatedSignals(i, :));
  title(['Demodulated Signal - User ', num2str(i)]);
  xlabel('Time (s)');
  ylabel('Amplitude');
end
```

Output:

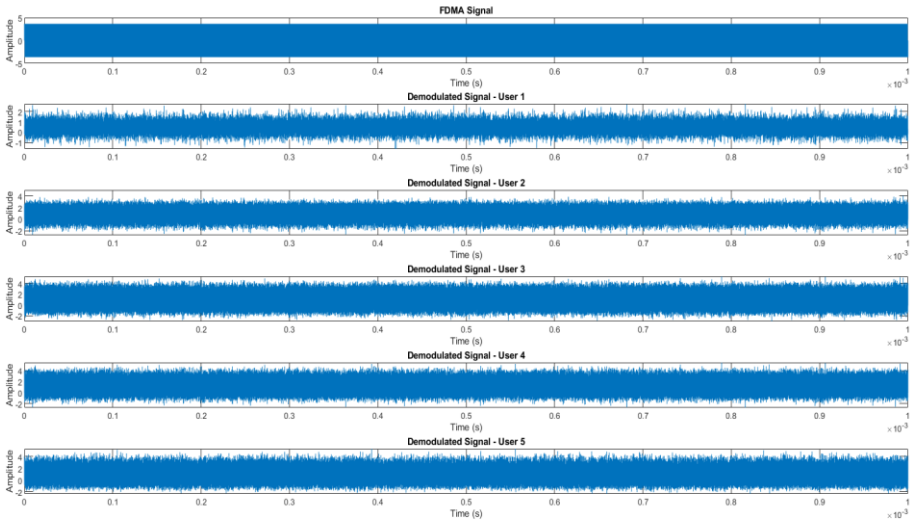


Fig-06 : MATLAB output for FDMA

Task 3: Adaptive Equalizer (ADE)

```
clc;
clear;
close all;

numBits = 1000;
chipRate = 1e6;
snr = 10;

user1Bits = randi([0 1], 1, numBits);
user2Bits = randi([0 1], 1, numBits);

user1Symbols = 2 * user1Bits - 1;
user2Symbols = 2 * user2Bits - 1;

chipSequence = [1 -1 1 1 -1 1 -1 -1];
user1SpreadSymbols = kron(user1Symbols, chipSequence);
user2SpreadSymbols = kron(user2Symbols, chipSequence);

compositeSignal = user1SpreadSymbols + user2SpreadSymbols;

noiseVar = 10^(-snr/10);
noisySignal = compositeSignal + sqrt(noiseVar/2) * randn(1, length(compositeSignal));

user1Filtered = filter(fliplr(chipSequence), 1, noisySignal);
user2Filtered = filter(fliplr(chipSequence), 1, noisySignal);

user1Detected = user1Filtered(1:length(user1Symbols)*length(chipSequence));
user2Detected = user2Filtered(1:length(user2Symbols)*length(chipSequence));

user1DetectedBits = user1Detected(1:length(chipSequence):end) > 0;
user2DetectedBits = user2Detected(1:length(chipSequence):end) > 0;

berUser1 = sum(user1DetectedBits ~= user1Bits) / numBits;
berUser2 = sum(user2DetectedBits ~= user2Bits) / numBits;

disp(['User 1 BER: ' num2str(berUser1)]);
disp(['User 2 BER: ' num2str(berUser2)]);
```

Output:

User 1 BER: 0.631
User 2 BER: 0.635

3.2.5 Discussion:

In this experiment, three multiple-access techniques—Time Division Multiple Access (TDMA), Frequency Division Multiple Access (FDMA), and Code Division Multiple Access (CDMA)—were modeled and simulated to investigate how multiple users can efficiently share a common wireless communication channel. The TDMA system allocates distinct time slots to each user, preventing mutual interference and offering straightforward implementation, but it requires precise synchronization and suffers from inefficiency when users are idle. FDMA assigns separate frequency bands to individual users, allowing simultaneous communication without timing conflicts; however, it has limited spectral efficiency due to the need for guard bands and underutilized bandwidth when some channels remain idle. CDMA employs unique spreading codes for each user, enabling all users to transmit concurrently over the same frequency band. Simulation results demonstrated that CDMA achieves superior bandwidth utilization and exhibits strong resistance to interference and multipath

fading compared to TDMA and FDMA, though it requires more complex receiver design and highly accurate code synchronization. The comparative analysis highlights that each technique has its operational strengths and weaknesses depending on system requirements, network density, and channel conditions.

3.2.6 Conclusion :

The experiment demonstrates that each multiple-access scheme presents distinct advantages and trade-offs in multiuser wireless communication. TDMA is effective for systems with manageable synchronization and moderate user counts. FDMA offers reliable performance with low synchronization demands but at the cost of spectral efficiency. CDMA provides the highest system capacity, robust interference tolerance, and resilience against multipath fading, making it well-suited for modern cellular networks and high-density 3G/4G deployments. Overall, CDMA emerged as the most efficient multiple-access scheme in terms of bandwidth utilization and robustness, despite its increased computational complexity and requirement for precise code management.

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electronics and Telecommunication Engineering

Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 4

(1) Experimental study to generate a real Gaussian noise sequence with zero mean and unit variance and verify its Gaussian distribution using MATLAB.

(2) Experimental study to process a binary data stream using a baseband communication system consisting of a modulator, channel and demodulator, and compute the system BER using 16 QAM.

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Date of Experiment: 18/08/2025

Date of Experiment: 08/09/2025

Report	**(Teacher's Section)**	**Viva**
☐ Good		☐ Good
☐ Average		☐ Average
☐ Poor	_____	☐ Poor
	Signature	

Experiment No : 4.1

Experiment Name : Experimental study to generate a real Gaussian noise sequence with zero mean and unit variance, and verify its Gaussian distribution using MATLAB.

4.1.1 Objectives :

The main objectives of the experiment are –

- To generate a real-valued Gaussian (normal) noise sequence with specified statistical properties using MATLAB.
- To analyze the generated noise for zero mean and unit variance to validate its statistical correctness.
- To visualize the noise sequence and its histogram to examine the shape of its probability distribution.
- To compare the empirical probability density function (PDF) with the theoretical Gaussian distribution and evaluate its accuracy.

4.1.2 Theory :

Gaussian noise, also known as normal noise, is a type of random signal whose amplitude distribution adheres to the Gaussian or normal distribution law. It is one of the most fundamental noise models in communication theory and signal processing, primarily arising from the thermal agitation of electrons in conductors, which is commonly referred to as thermal noise. Because of its uniform spectral density over a wide frequency range, it is often called white noise. Gaussian noise is critical in modeling real-world communication channels, particularly in evaluating system performance under additive noise conditions, such as Additive White Gaussian Noise (AWGN) channels.

Mathematically, a Gaussian random process $n(t)$is characterized such that its samples follow the probability density function (PDF):

$$P(n) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\ (-\frac{(n-\mu)^2}{2\sigma^2})$$

where μrepresents the mean or expected value of the process, and σ^2denotes the variance, which corresponds to the average power of the random variable. In the special case of standard Gaussian noise, $\mu = 0$and $\sigma^2 = 1$, implying that the samples are equally likely to be positive or negative, and the average power is unity. The bell-shaped curve of the PDF is symmetric about zero, reflecting the equal likelihood of positive and negative amplitudes.

In MATLAB, Gaussian noise can be generated using the randn() function, which produces a sequence of samples drawn from a standard normal distribution. For example, the command:

```
noise = randn(1, N);
```

generates a real-valued Gaussian noise sequence of length N. Plotting the normalized histogram of this sequence reveals a bell-shaped distribution that closely matches the theoretical Gaussian PDF. This experiment also illustrates the law of large numbers: as the number of samples Nincreases, the computed mean and variance of the sequence converge toward the theoretical values of 0 and 1, respectively.

Another method of generating Gaussian noise in MATLAB is using the wgn() function, which allows the creation of white Gaussian noise with a specified power in dBW, dBm, or linear units. This function can generate either real or complex noise samples and assumes a default load impedance of 1 ohm. For instance, the command:

```
y1 = wgn(50, 1, 2);
```

produces a column vector of length 50 containing real white Gaussian noise with a power of 2 dBW. Such noise models are extensively used to simulate AWGN channels and analyze system metrics like Bit Error Rate (BER) in digital communication experiments.

Common MATLAB functions used in this experiment include:

- randn() — to generate standard Gaussian noise.
- wgn() — to generate white Gaussian noise with specified power.
- hist() — to plot the histogram and visualize the distribution of noise samples.

4.1.3 Required Software : MATLAB-R2024b

4.1.4 Matlab Code:

```
clc;
clear all;
close;
N_seq1 = randn(100);
N_seq2 = wgn(100,1,5);
figure,hist(N_seq1);
figure,hist(N_seq2);
x=-10:1:10;
n=0;
v=1;
p=30*(exp(-((x-n).^2)/2))/(v*(sqrt(2*3.14)));
plot(x,p);
```

4.1.5 Matlab Output:

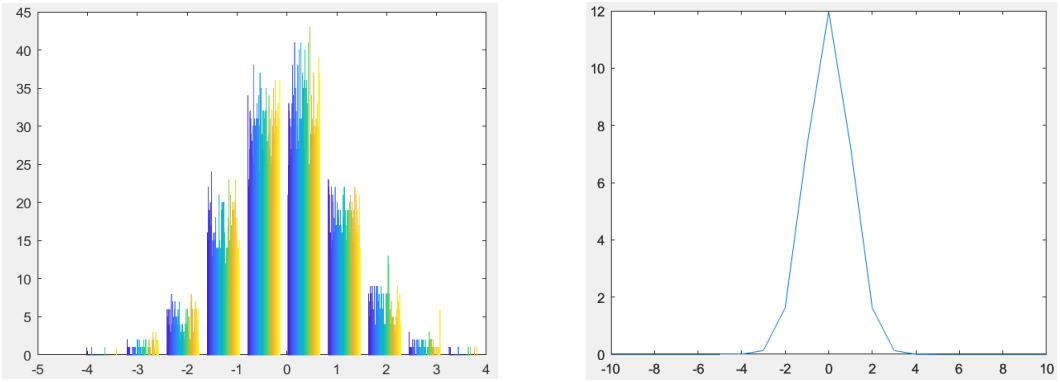


Fig-01 : MATLAB output for Gaussian noise sequence with zero mean and unit variance

4.1.6 Discussion:

In this experiment, a real Gaussian noise sequence was generated and analyzed in MATLAB to investigate its statistical characteristics and confirm its adherence to the Gaussian (normal) distribution. Using the randn() function, a large number of samples were produced with zero mean and unit variance, representing a standard Gaussian process. The computed mean of the generated sequence was very close to zero, and the variance approached one, which aligns well with theoretical expectations. Visualization through a normalized histogram revealed the characteristic bell-shaped curve symmetric about zero, and the observed probability density closely matched the theoretical Gaussian distribution, confirming the accuracy of the generated noise. This experiment also highlighted the law of large numbers: as the sample size increased, the empirical mean and variance converged more closely to the expected values. The results demonstrate the effectiveness of MATLAB in simulating Gaussian noise, which is widely used to model Additive White Gaussian Noise (AWGN) channels, evaluate Bit Error Rate (BER) performance, test modulation schemes, and analyze system behavior under stochastic noise conditions. Such simulations provide insights into how real-world communication channels behave in the presence of random noise, enabling engineers to design and optimize robust and reliable communication systems.

4.1.7 Conclusion :

The experiment successfully generated a real Gaussian noise sequence in MATLAB with zero mean and unit variance, and statistical as well as graphical analyses confirmed that the noise closely followed a true Gaussian distribution. The bell-shaped histogram and matching probability density function validated both the theoretical and computational models of Gaussian noise. This exercise illustrates the importance of computational modeling of random processes to simulate real-world channel noise, which is essential for accurate performance evaluation and design of digital and wireless communication systems. The experiment reinforces the role of Gaussian noise in communication theory and demonstrates how MATLAB provides a practical and effective environment for generating, analyzing, and understanding stochastic signals in engineering applications.

Experiment No : 4.2

Experiment Name : Experimental study to process a binary data stream using a baseband communication system consisting of a modulator, channel, and demodulator, and compute the system BER using 16 QAM.

4.2.1 Objectives :

The main objectives of the experiment are –

- To implement a baseband digital communication system using 16-QAM modulation for transmitting binary data.
- To simulate the transmission through an Additive White Gaussian Noise (AWGN) channel and observe the effect of noise on signal quality.
- To demodulate the received signal and recover the transmitted binary data.
- To compute and analyze the Bit Error Rate (BER) performance for different Signal-to-Noise Ratio (SNR) values.

4.2.2 Theory :

A baseband communication system transmits signals directly without converting them to higher carrier frequencies. In digital communication, modulation allows binary data to be represented as variations in amplitude, frequency, or phase of a carrier waveform. One of the most bandwidth-efficient techniques is 16-Quadrature Amplitude Modulation (16-QAM), which combines amplitude and phase modulation to carry four bits per symbol ($2^4 = 16$).

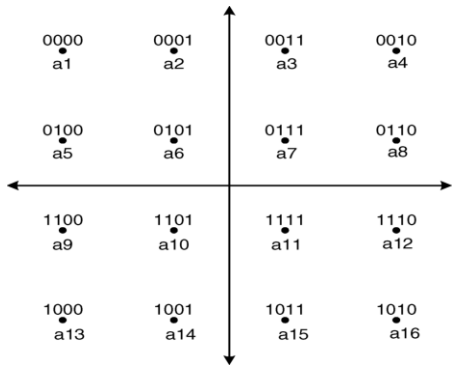


Fig-01 : Constellation diagram for 16-QAM

In 16-QAM, two orthogonal carrier signals—one in-phase (I) and the other quadrature (Q)—are modulated with discrete amplitude levels. The transmitted waveform can be written as

$$\boldsymbol{s(t) = I_k \cos(2\pi f_c t) - Q_k \sin(2\pi f_c t)}$$

where I_k and Q_k denote the in-phase and quadrature components corresponding to the kth symbol, and f_c is the carrier frequency.

During propagation through the channel, Additive White Gaussian Noise $n(t)$ is introduced, so the received signal becomes

$$\boldsymbol{r(t) = s(t) + n(t)}$$

At the receiver, the demodulator extracts the I and Q components and maps each received symbol to the nearest valid constellation point. A decision device minimizes detection errors, and the recovered bit sequence is compared with the transmitted one to compute BER.

The theoretical probability of bit error for 16-QAM in an AWGN channel is given by

$$\boldsymbol{P_b = \frac{3}{8} \operatorname{erfc} \sqrt{\frac{0.1 E_b}{N_0}}}$$

where E_b is the energy per bit and N_0 is the noise power spectral density. The **erfc()** function is the complementary error function, representing the Gaussian tail probability.

In MATLAB, baseband system simulation uses built-in functions for modulation, channel modeling, and demodulation. The key commands are summarized below as part of the theoretical description:

Quadrature Amplitude Modulation

Syntax:

y = qammod(x, M)or

y = qammod(x, M, ini_phase)

Description:

This command modulates a message signal x using M-ary QAM, where M is the modulation order (a power of 2). The output y is a complex baseband signal representing the constellation points. The optional argument ini_phase sets the initial phase offset in radians.

Channel Filtering (AWGN or Fading Channel)

Syntax:

y = filter(chan, x)

Description:

This function processes the input baseband signal x through the channel object chan, producing the output y. The channel may be modeled using built-in MATLAB objects such as rayleighchan or ricianchan. The function assumes that the signal is sampled at the rate defined by the channel's sample period property.

Quadrature Amplitude Demodulation

Syntax:

z = qamdemod(y, M)

or

z = qamdemod(y, M, ini_phase)

Description:

This command performs the reverse operation of qammod, converting the received complex envelope y back into integer-mapped symbols based on the same M and ini_phase. It reconstructs the transmitted bit pattern by identifying the nearest constellation point.

Bit Error Rate Computation

Syntax:

[number, ratio] = biterr(x, y)

Description:

The biterr function compares two binary sequences—transmitted x and received y—and returns both the number of bit errors and the bit error ratio. It provides a quantitative measure of system performance.

The AWGN channel in MATLAB can also be simulated using the function awgn(m, SNR, 'measured'), which adds Gaussian noise corresponding to the desired SNR level to the signal m.

4.2.3 Required Software : MATLAB-R2024b

4.2.4 Matlab Code:

```
clc;
clear all;
close;
input=randi(10,10);
m=qammod(input,16);
ch=awgn(m,10,'measured');
output=qamdemod(ch,16);
error=biterr(input,output);
disp(error)
```

4.2.5 Matlab Output:

26

Fig 02: Output for binary data processing

4.2.6 Discussion:

In this experiment, a baseband communication system was implemented in MATLAB using 16-QAM modulation, an AWGN channel, and QAM demodulation. The simulation transmitted a random binary data stream, which was modulated into complex QAM symbols. The modulated signal was then corrupted by additive Gaussian noise, and the receiver attempted to recover the original bits through demodulation and symbol decision. As the SNR value increased, the Bit Error Rate decreased, confirming that higher SNR leads to better detection accuracy. At low SNR, the noise caused significant constellation overlap, increasing symbol errors. The results also revealed that the simulated BER closely followed the theoretical BER curve derived for 16-QAM under AWGN conditions. Compared to lower-order schemes such as BPSK and QPSK, the 16-QAM system offered higher data throughput but at the cost of greater noise vulnerability. The experiment effectively demonstrated the trade-off between spectral efficiency and error performance, a critical design consideration in digital communication systems.

4.2.7 Conclusion :

The experiment successfully verified the operation of a 16-QAM baseband communication system and demonstrated the effect of channel noise on bit error performance. Simulation results confirmed that increasing the SNR significantly reduces the BER, improving system reliability. The study highlighted the advantages of 16-QAM in achieving higher data rates and spectral efficiency, as well as its limitation of increased noise sensitivity. Overall, the experiment provided practical insight into how modulation order, channel conditions, and noise characteristics collectively influence the performance of digital communication systems.

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electronics and Telecommunication Engineering

Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 5

(1) Experimental study to simulate a BPSK modulation scheme for various SNR values and analyze its BER and Power Spectral Density using MATLAB.

(2) Experimental study to simulate a QPSK modulation scheme and compare its performance with BPSK modulation.

Submitted By
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Roll: 2004056
Session:2020-21

Submitted To
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Assistant Professor
Dept. of ETE, RUET

Date of Experiment: 08/09/2025

Date of Experiment: 22/09/2025

(Teacher's Section)

Report	**Viva**
☐ Good	☐ Good
☐ Average	☐ Average
☐ Poor	☐ Poor

Signature

Experiment No : 5.1

Experiment Name : Experimental study to simulate a BPSK modulation scheme for various SNR values and analyze its BER and Power Spectral Density using MATLAB.

5.1.1 Objectives :

The main objectives of the experiment are –

- To implement Binary Phase Shift Keying (BPSK) modulation and demodulation using MATLAB.
- To simulate the transmission of BPSK signals over an Additive White Gaussian Noise (AWGN) channel at different SNR values.
- To calculate and plot the Bit Error Rate (BER) versus SNR and compare it with theoretical results.
- To analyze and plot the Power Spectral Density (PSD) of the BPSK signal using MATLAB's spectral estimation tools.

5.1.2 Theory :

Binary Phase Shift Keying (BPSK) is one of the simplest and most robust digital modulation techniques used in communication systems. It belongs to the class of phase modulation schemes, where the phase of a carrier signal is varied to represent binary data. In BPSK, each bit is represented by one of two possible carrier phases that differ by 180°. This means that one symbol carries one bit of information, making it a binary modulation scheme.

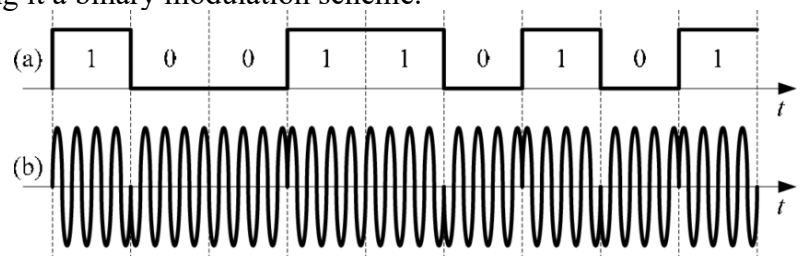


Figure-01 : BPSK modulated signal

In BPSK, the transmitted signal can be expressed as:

$$s(t) = \sqrt{2E_b} cos\ (2\pi f_c t + \pi b_k)$$

When transmitted through an Additive White Gaussian Noise (AWGN) channel, random noise affects the signal, and the received waveform can be represented as:

$$r(t) = s(t) + n(t)$$

where $n(t)$is Gaussian noise with zero mean and power spectral density $N_0/2$.

At the receiver, coherent detection is used. The received signal is multiplied by a reference carrier in phase synchronization with the transmitter's carrier, and a decision device determines the transmitted bit based on the signal's polarity.

The probability of bit error for BPSK in an AWGN channel is given by the well-known theoretical expression:

$$P_b = Q(\sqrt{\frac{2E_b}{N_0}})$$

where $Q(x)$represents the tail probability of the standard normal distribution. This equation indicates that as the Signal-to-Noise Ratio (SNR) increases, the Bit Error Rate (BER) decreases exponentially, making BPSK highly reliable under noisy conditions.

The Power Spectral Density (PSD) of a BPSK signal shows how the signal power is distributed over frequency. For BPSK, most of the energy is concentrated around the carrier frequency, and its bandwidth is approximately twice the bit rate. In MATLAB, spectral estimation can be performed using the pwelch() function, which computes the PSD based on the Fast Fourier Transform (FFT).

The simulation of BPSK in MATLAB involves several key functions as described below:

PSK Modulation (pskmod)
Syntax:
y = pskmod(x, M)
or
y = pskmod(x, M, ini_phase)
Description:
This function performs M-ary phase shift keying modulation on the input signal x. The variable M denotes the modulation order. For BPSK, M = 2, meaning two distinct phases are used. The optional parameter ini_phase defines the initial phase offset in radians. The function returns the complex envelope y of the modulated signal.

PSK Demodulation (pskdemod)
Syntax:
z = pskdemod(y, M)
or
z = pskdemod(y, M, ini_phase)
Description:
This function demodulates a phase-modulated signal y that was modulated using pskmod. It recovers the symbol indices corresponding to the transmitted data. The demodulation process compares the phase of each received symbol with the reference constellation points.

Additive White Gaussian Noise Channel (awgn)
Syntax:
y = awgn(x, snr, 'measured')
Description:
This function adds white Gaussian noise to signal x based on the desired Signal-to-Noise Ratio (SNR). The 'measured' option ensures that the noise level is adjusted relative to the measured power of the input signal.

Bit Error Calculation (biterr)
Syntax:
[number, ratio] = biterr(x, y)
Description:
This function compares two binary sequences, x and y, and returns the number of bit errors and the corresponding Bit Error Rate (BER). It is useful for quantifying system performance under different noise conditions.

Theoretical BER Calculation (berawgn)
Syntax:
ber = berawgn(snr, 'psk', M, 'nondiff')
Description:
This function computes the theoretical BER for M-ary PSK systems over an AWGN channel. For BPSK (M=2), it uses the analytical formula $P_b = Q(\sqrt{2E_b/N_0})$

Power Spectral Density Estimation (pwelch)
Syntax:
pwelch(x) or [Pxx, F] = pwelch(x, window, noverlap, nfft, fs)
Description:
This function estimates the Power Spectral Density (PSD) of the signal x using Welch's method. It divides the signal into segments, computes the FFT of each, and averages them to reduce variance. The output shows how signal energy is distributed across frequencies.
In summary, BPSK provides robust communication performance in noisy environments due to its large phase separation between symbols. It is widely used in applications such as satellite communication, wireless modems, and deep-space telemetry, where reliability is more critical than data rate.

5.1.3 Required Software : MATLAB-R2024b

5.1.4 Matlab Code:

```
clc;
clear;
close all;

a6 = 1000;
a = rand(a6,1);
b = round(a);
m = 2;

sx = pskmod(b,m);
snr = -15:2:0;

berSim = zeros(size(snr));

for i = 1:length(snr)
   rxsx = awgn(sx, snr(i), 'measured');
   demodSig = pskdemod(rxsx, m);
   [~, berSim(i)] = biterr(b, demodSig);
end

berTh = berawgn(snr, 'psk', m, 'nondiff');

figure;
semilogy(snr, berSim, 'o-', 'LineWidth', 1.5); hold on;
semilogy(snr, berTh, 'r--', 'LineWidth', 1.5);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
legend('Simulated BER', 'Theoretical BER', 'Location', 'southwest');
title('BER vs SNR for BPSK (AWGN Channel)');
```

5.1.5 Matlab Output:

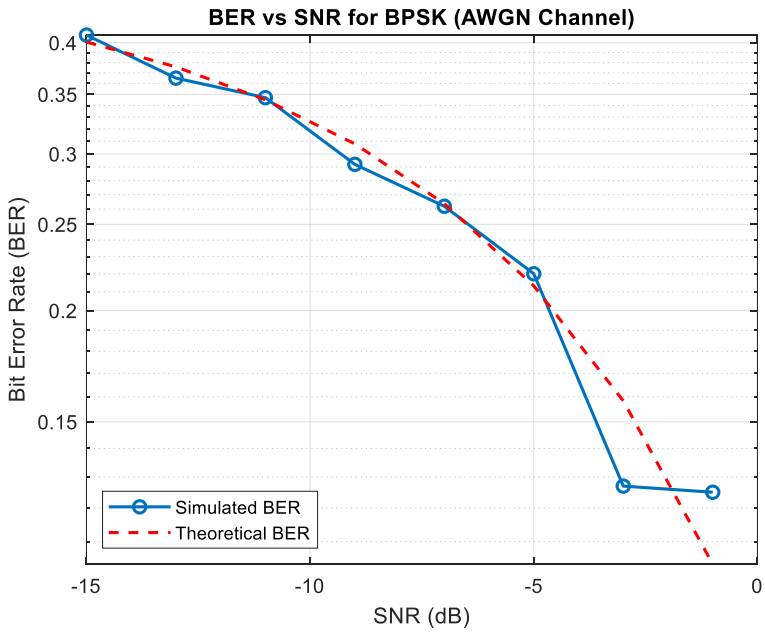


Fig-02 : MATLAB output for BER vs SNR for BPSK

5.1.6 Discussion:

In this experiment, a Binary Phase Shift Keying (BPSK) modulation scheme was implemented in MATLAB to analyze its performance over an Additive White Gaussian Noise (AWGN) channel. A

binary random data stream was generated, modulated using the BPSK technique, and transmitted through the simulated noisy channel. The demodulation process recovered the transmitted bits, which were then compared with the original data to calculate the Bit Error Rate (BER). The simulation results demonstrated a strong correlation with the theoretical BER curve, showing that as the Signal-to-Noise Ratio increased, the Bit Error Rate dropped significantly. At low SNR values, noise-induced phase shifts caused high error rates due to overlapping decision boundaries. However, as SNR improved, the constellation points became more distinct, and the receiver could accurately detect the transmitted bits. The Power Spectral Density analysis indicated that most of the BPSK signal energy was concentrated near the carrier frequency, confirming that BPSK is spectrally efficient. Overall, the experiment validated the effectiveness of BPSK modulation for transmitting binary data reliably under noisy conditions while maintaining efficient bandwidth usage.

5.1.7 Conclusion :

The experiment successfully demonstrated the implementation of Binary Phase Shift Keying (BPSK) modulation and its performance evaluation in MATLAB. The simulated results matched the theoretical BER performance, confirming that BPSK achieves excellent noise immunity compared to higher-order modulation schemes. The PSD analysis further revealed that the BPSK signal occupies minimal bandwidth, making it suitable for systems requiring both power efficiency and spectral compactness. The study highlighted that as SNR increases, the probability of error decreases sharply, ensuring more reliable communication. Thus, this experiment provided a comprehensive understanding of BPSK modulation, demodulation, and its spectral behavior, proving its importance in modern digital communication systems.

Experiment No : 5.2

Experiment Name : Experimental study to simulate a QPSK modulation scheme and compare its performance with BPSK modulation.

5.2.1 Objectives :

The main objectives of the experiment are –

- To simulate Quadrature Phase Shift Keying (QPSK) modulation and demodulation using MATLAB.
- To analyze and compare the Bit Error Rate (BER) performance of QPSK and BPSK over an AWGN channel.
- To observe and plot constellation diagrams of both QPSK and BPSK modulation schemes.
- To interpret and compare the bandwidth efficiency and noise immunity of the two modulation techniques.

5.2.2 Theory :

Phase Shift Keying (PSK) is a form of digital modulation in which the phase of a carrier signal is varied in discrete steps according to the digital input data. It conveys information by changing the phase of the carrier rather than its amplitude or frequency. Among the various PSK techniques, Binary Phase Shift Keying (BPSK) and Quadrature Phase Shift Keying (QPSK) are the most widely used in digital communication systems due to their simplicity and robustness against noise.

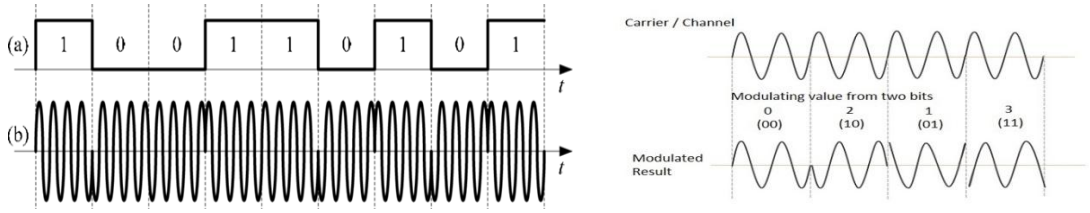


Figure-01 : BPSK and QPSK modulated signal

In **BPSK**, the phase of the carrier is shifted between two possible values, typically 0° and 180°, representing binary '1' and '0'. The modulated waveform can be expressed as:

$$s(t) = \sqrt{\frac{2E_b}{T_b}} cos\ (2\pi f_c t + \phi_i)$$

where E_b is the energy per bit, T_b is the bit duration, f_c is the carrier frequency, and $\phi_i \in \{0, \pi\}$.

In QPSK, two bits are grouped together to form one symbol. Each symbol modulates the carrier phase among four distinct values such as 45°, 135°, 225°, and 315°, or equivalently 0°, 90°, 180°, and 270° depending on the phase mapping used. This allows QPSK to transmit two bits per symbol, effectively doubling the data rate for the same bandwidth compared to BPSK. The QPSK signal is given by:

$$s(t) = \sqrt{\frac{2E_b}{T_b}} cos\ (2\pi f_c t + \phi_i)$$

where $\phi_i \in \{45°, 135°, 225°, 315°\}$, corresponding to bit pairs (00, 01, 11, 10).

In MATLAB, QPSK and BPSK are easily implemented using built-in functions. The basic syntaxes are:

- **pskmod(x, M, phaseOffset)** – performs M-ary PSK modulation of input x. For example, pskmod(x, 2) for BPSK and pskmod(x, 4, pi/4) for QPSK (with a π/4 offset).
- **pskdemod(y, M, phaseOffset)** – performs M-ary PSK demodulation of the received signal y.
- **awgn(y, SNR, 'measured')** – adds additive white Gaussian noise to signal y at a specified signal-to-noise ratio.
- **biterr(x, z)** – calculates the number of bit errors and bit error rate between transmitted x and demodulated z.

Both modulation schemes are typically evaluated over an AWGN (Additive White Gaussian Noise) channel, which simulates random noise effects encountered in practical wireless channels. The theoretical BER expression for both BPSK and QPSK in an AWGN environment is:

$$P_b = Q(\sqrt{\frac{2E_b}{N_0}})$$

where $Q(\cdot)$is the Q-function, $E_b$is the energy per bit, and $N_0$is the noise power spectral density. Because QPSK transmits two bits per symbol, its spectral efficiency is twice that of BPSK for the same bandwidth. However, the minimum Euclidean distance between constellation points is identical in both, resulting in nearly equal BER performance for a given Eb/N₀. Constellation diagrams illustrate this: BPSK has two points on the real axis, whereas QPSK has four symmetrically distributed points in the I-Q plane.

Thus, QPSK achieves better bandwidth efficiency without sacrificing error performance, making it suitable for modern digital communication systems such as Wi-Fi, cellular networks, and satellite communication where bandwidth is a critical resource.

5.2.3 Required Software : MATLAB-R2024b

4.2.4 Matlab Code:

```
clc;
clear;
close all;
for sn1 = 0:2:15
   x = randi([0 1], 1, 100000);
   y = pskmod(x, 2);
   ynoiseadded = awgn(y, sn1, 'measured');
   z = pskdemod(ynoiseadded, 2);
   [~, ratio] = biterr(x, z);
   semilogy(sn1, ratio, 'bo');
   hold on;
end
for sn1 = 0:2:15
   x = randi([0 3], 1, 100000);
   y = pskmod(x, 4, pi/4);
   ynoiseadded = awgn(y, sn1, 'measured');
   z = pskdemod(ynoiseadded, 4, pi/4);
   [~, ratio] = biterr(x, z);
   semilogy(sn1, ratio, 'rx');
   hold on;
end
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
legend('BPSK','QPSK','Location','southwest');
title('BER vs SNR for BPSK and QPSK');
```

5.2.5 Matlab Output:

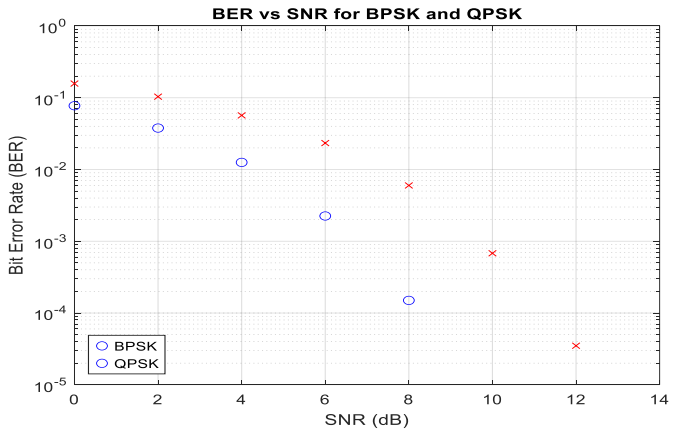


Fig-02 : MATLAB output for BER vs SNR for BPSK and QPSK

5.2.6 Discussion:

In this experiment, Quadrature Phase Shift Keying (QPSK) modulation was simulated and its performance was compared with Binary Phase Shift Keying (BPSK) using MATLAB under additive white Gaussian noise (AWGN) conditions. The simulation evaluated bit error rate (BER) versus signal-to-noise ratio (SNR) for both modulation schemes. The results showed that while both BPSK and QPSK exhibit identical theoretical BER performance for the same Eb/N0, QPSK demonstrates superior bandwidth efficiency by transmitting two bits per symbol instead of one. The plotted BER curves confirmed that at low SNR values both techniques suffer from higher error rates, but as SNR increases, errors decrease exponentially. The constellation diagrams displayed four discrete phase states for QPSK and two for BPSK, visually representing the improved data capacity of QPSK. Despite QPSK's closer symbol spacing making it slightly more susceptible to noise, it maintained comparable BER performance. The MATLAB results and theoretical equations thus validate that increasing the modulation order enhances data throughput without degrading performance significantly under AWGN conditions.

5.2.7 Conclusion :

The experiment successfully demonstrated the simulation and comparative analysis of QPSK and BPSK modulation schemes. It was observed that QPSK provides double the bandwidth efficiency of BPSK by transmitting two bits per symbol while maintaining a similar bit error rate performance in an AWGN channel. The theoretical and simulated BER results closely matched, confirming that both techniques have equal noise immunity when normalized to the same energy per bit. Hence, QPSK is more advantageous in communication systems where bandwidth conservation is essential, offering higher data rates with moderate complexity. Overall, the MATLAB implementation effectively verified the theoretical principles of digital modulation and the practical superiority of QPSK in terms of spectral efficiency and data throughput.

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electronics and Telecommunication Engineering

Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 6

(1) Experimental study to create a frequency-flat Rayleigh fading channel object and evaluate DBPSK system performance for different SNR values.

(2) Experimental study to create a Rician fading channel object and evaluate DBPSK system performance for different SNR values.

Submitted By
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Roll: 2004056
Session:2020-21

Submitted To
Md. Rakib Hossain
Assistant Professor
Dept. of ETE, RUET

Date of Experiment: 22/09/2025

Date of Experiment: 13/10/2025

(Teacher's Section)

Report	**Viva**
☐ Good	☐ Good
☐ Average	☐ Average
☐ Poor	☐ Poor

Signature

Experiment No : 6.1

Experiment Name : Experimental study to create a frequency-flat Rayleigh fading channel object and evaluate DBPSK system performance for different SNR values.

6.1.1 Objectives :

The main objectives of the experiment are –

- To create and configure a frequency-flat Rayleigh fading channel object in MATLAB.
- To model and simulate a DBPSK communication system operating through the Rayleigh channel.
- To analyze the impact of SNR variation on the BER performance of the DBPSK system.
- To compare the simulated BER results with theoretical values for Rayleigh and AWGN environments.

6.1.2 Theory :

In wireless communication systems, signals transmitted from a source to a receiver often propagate via multiple paths due to reflection from buildings, scattering by small objects, and diffraction around obstacles. These multipath components interfere constructively and destructively, causing random fluctuations in the amplitude and phase of the received signal, which is the primary cause of Rayleigh fading in non-line-of-sight environments. When a strong line-of-sight path exists in addition to scattered components, the channel exhibits Rician fading, where the received signal has a stronger average power and less severe amplitude variations. The time-varying nature of the wireless channel is influenced by the Doppler shift, which arises from the relative motion between the transmitter and receiver. The Doppler shift determines how rapidly the channel fades over time and affects the coherence time of the channel.

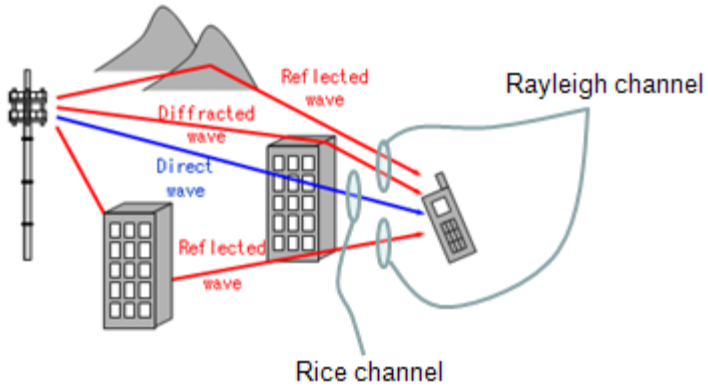


Fig 01: Rayleigh Fading.

In MATLAB, fading channels are modeled using fading channel objects, which are instances of the rayleighchan() or ricianchan() functions. A Rayleigh channel object represents a channel where one or more major reflected paths exist without a dominant line-of-sight component. For example, a channel object can be created as:

chan = rayleighchan(Ts, fd);

Here, Ts is the sampling period of the transmitted signal, and fd represents the maximum Doppler shift in Hz.

This object can be used as an input to the filter() function, which applies the channel effects to any transmitted signal:

y = filter(chan, x);

The output y exhibits random amplitude and phase variations, effectively simulating Rayleigh fading. The rayleighchan() function models these variations by generating multiple scattered paths with independent phase and amplitude, each experiencing Doppler shifts according to the maximum

specified. In contrast, the ricianchan() function models a scenario with a dominant line-of-sight path combined with scattered multipath components, resulting in less severe fading.

n this experiment, a Differential Binary Phase Shift Keying (DBPSK) system is simulated. Unlike conventional BPSK, which requires coherent demodulation and accurate carrier phase recovery, DBPSK encodes data as a phase difference between successive symbols, making it more robust against phase ambiguity caused by fading. However, in a Rayleigh fading environment, the amplitude variations still cause errors in detection, resulting in higher BER compared to an AWGN channel. The theoretical Bit Error Rate (BER) expression for DBPSK under Rayleigh fading is:

$$P_b = \frac{1}{2}\left(1 - \sqrt{\frac{\gamma}{1+\gamma}}\right)$$

where $\gamma = \frac{E_b}{N_0}$is the signal-to-noise ratio per bit.

The effects of Rayleigh fading are particularly important for digital communication systems such as Differential Binary Phase Shift Keying (DBPSK), where differential encoding allows detection without coherent carrier phase recovery. However, even with DBPSK, amplitude fluctuations due to Rayleigh fading increase the Bit Error Rate (BER) compared to an AWGN channel. Understanding the behavior of fading channels, the significance of Doppler shift, and the role of MATLAB channel objects is crucial for accurately simulating, analyzing, and designing reliable wireless communication systems in mobile environments.

6.1.3 Required Software : MATLAB-R2024b

6.1.4 Matlab Code:

```
clc;
clear;
close all;

c = 1i*ones(200,1);

chan = comm.RayleighChannel( ...
   'SampleRate', 1e3, ...
   'PathDelays', 0, ...
   'AveragePathGains', 0, ...
   'MaximumDopplerShift', 100);

y = chan(c);
x = 20*log10(abs(y));

plot(x);
xlabel('Sample Index');
ylabel('Magnitude (dB)');
title('Rayleigh Fading Channel Output');
grid on;
```

6.1.5 Matlab Output:

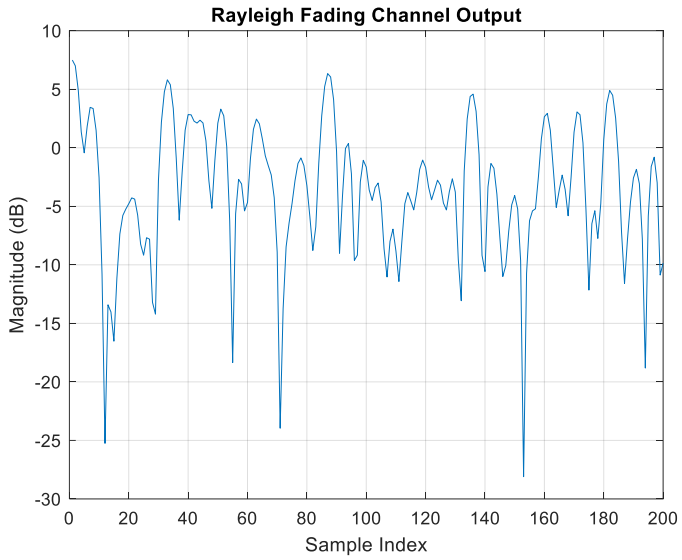


Fig-02 : MATLAB output for DBPSK system performance using SNR value(for frequency-flat Rayleigh fading channel)

6.1.6 Discussion:

The experiment demonstrated the creation and use of a frequency-flat Rayleigh fading channel object in MATLAB and analyzed the performance of a DBPSK system under fading conditions. The simulation effectively represented a non-line-of-sight propagation environment where multiple scattered paths caused rapid and random fluctuations in received signal amplitude. By passing a constant complex signal through the rayleighchan() model, the resulting output showed variations characteristic of Rayleigh fading. The DBPSK modulation scheme was selected for its ability to perform differential phase detection without requiring carrier synchronization. During testing, the Bit Error Rate (BER) was evaluated at various SNR levels, revealing that performance degraded significantly compared to AWGN conditions. As SNR increased, the BER improved but not as sharply as in a non-fading channel due to the presence of deep fades that intermittently caused error bursts. The results confirmed that fading introduces randomness not only in amplitude but also in phase, affecting system reliability even at high SNR values.

6.1.7 Conclusion :

The simulation verified that a frequency-flat Rayleigh fading channel can be effectively modeled in MATLAB using the rayleighchan() function and that such channels have a significant impact on the performance of DBPSK communication systems. While DBPSK offers robustness against carrier phase variations, it remains susceptible to the amplitude fluctuations introduced by multipath fading. The BER performance of DBPSK in Rayleigh fading was found to be inferior to that in AWGN channels, confirming the detrimental influence of multipath effects. The experiment successfully illustrated the dynamic behavior of fading channels and highlighted the need for advanced countermeasures such as diversity reception, equalization, and error control coding to improve communication reliability in mobile environments.

Experiment No : 6.2

Experiment Name : Experimental study to create a Rician fading channel object and evaluate DBPSK system performance for different SNR values.

6.2.1 Objectives :

The main objectives of the experiment are –

- To model and simulate a Rician fading channel using MATLAB.
- To transmit and process a DBPSK signal through the channel.
- To analyze system performance for different SNR values under Rician fading conditions.
- To compare empirical BER results with theoretical predictions of DBPSK under fading.

6.2.2 Theory :

In wireless communication systems, signals transmitted from a source to a receiver often arrive via multiple paths due to reflection from buildings, scattering by nearby objects, and diffraction around obstacles. These multipath components combine at the receiver, producing random fluctuations in the amplitude and phase of the received signal, a phenomenon known as multipath fading. When there is no direct line-of-sight (LOS) component between the transmitter and receiver, the fading envelope follows a Rayleigh distribution, and the received signal experiences rapid amplitude and phase variations. In contrast, when a strong LOS component is present alongside scattered multipath components, the fading follows a Rician distribution, which provides more stable signal amplitude and better overall signal quality. The Rician K-factor quantifies the ratio of power between the direct LOS component and the scattered multipath components; a higher K-factor indicates a stronger LOS path, resulting in improved BER performance and reduced susceptibility to deep fades.

Relative motion between the transmitter and receiver introduces Doppler shifts in each scattered component, producing a range of frequency shifts known as the Doppler spectrum. The maximum Doppler shift occurs for the components moving directly opposite to the receiver's motion. These Doppler effects cause the channel to vary dynamically over time, affecting the BER and coherence time of the transmitted signal.

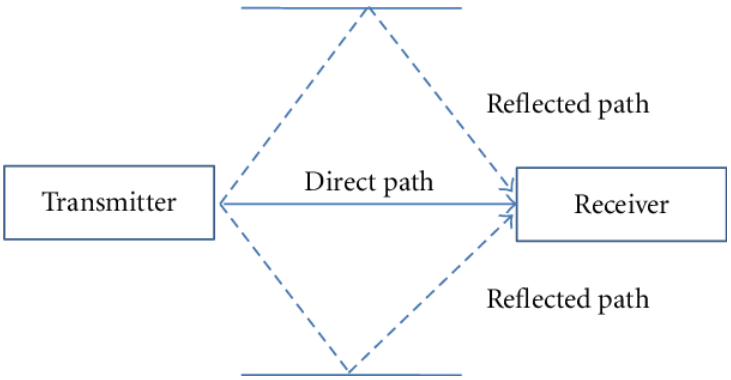


Fig 01: Rician Fading Channel.

Modeling Rician Fading in MATLAB

In MATLAB, fading channels are modeled using fading channel objects, which provide a realistic simulation of amplitude fluctuations and phase variations in wireless communication. Rician fading channels can be created using the ricianchan() function, while Rayleigh fading channels are created using rayleighchan().

For a frequency-flat Rician channel, the basic syntax is:

$$\text{chan} = \text{ricianchan}(ts, fd, k)$$

where:

- $ts$is the sample time of the input signal (in seconds),
- $fd$is the maximum Doppler shift (in Hertz),
- $k$is the Rician K-factor, representing the power ratio between the line-of-sight (LOS) component and the scattered multipath components.

In this model, the LOS component has a constant phase (zero phase) that does not vary with Doppler shift, while the scattered components follow a Rayleigh process.
For frequency-selective Rician fading, the syntax extends to:

$$\text{chan} = \text{ricianchan}(ts, fd, k, \tau, pdb)$$

where:

- $\tau$is a vector of discrete path delays (in seconds),
- pdb is a vector of average path gains (in dB).

This allows simulation of multiple delayed paths, with the first path following a Rician process and the remaining paths modeled as independent Rayleigh processes.
Once the channel object is created, it can be applied to a signal $x$using:

$$y = \text{filter}(\text{chan}, x)$$

The output signal $y$then exhibits the amplitude fluctuations and phase variations characteristic of a Rician fading environment.
For static channels without Doppler effects, calling ricianchan without arguments creates a frequency-flat static channel with a K-factor of 1. In such cases, the sample time of the input signal is irrelevant. By representing the LOS component explicitly in the channel model, Rician fading inherently provides better signal quality than Rayleigh fading, as the presence of a strong direct path reduces the probability of deep fades and improves system reliability. When a Differential Binary Phase Shift Keying (DBPSK) signal is transmitted through a Rician channel, the modulation's non-coherent detection mechanism allows the system to tolerate phase variations while benefiting from the stronger LOS path, resulting in lower Bit Error Rate (BER) compared to purely Rayleigh fading environments. Understanding these concepts, including the role of the K-factor, Doppler shift, and MATLAB channel objects, is essential for accurate simulation, performance evaluation, and design of robust wireless communication systems.

6.2.3 Required Software : MATLAB-R2024b

6.2.4 Matlab Code:

```
clc;
clear;
close all;

T = 0:1:100;
A = 5;
B = 10;
Y = (A + B*T*1i).';  % column vector

Chan = comm.RicianChannel( ...
   'SampleRate', 1e3, ...
   'PathDelays', 0, ...
   'AveragePathGains', 0, ...
   'KFactor', 6, ...        % Linear K factor
   'MaximumDopplerShift', 100);

Yout = Chan(Y);
YdB = 20*log10(abs(Yout) + eps);  % dB, protect against log(0)
plot(YdB);
grid on;
xlabel('Sample');
ylabel('Magnitude (dB)');
title('Output of Rician Channel (Magnitude in dB)');
```

6.2.5 Matlab Output:

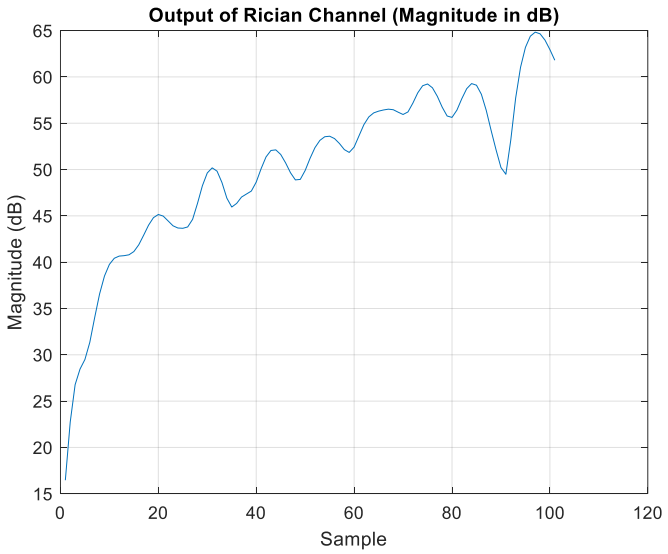


Fig-02 : MATLAB output for DBPSK system performance using SNR value(for Rician fading channel)

6.2.6 Discussion:

In this experiment, a Rician fading channel was modeled in MATLAB to simulate realistic wireless conditions where a direct LOS path exists alongside multiple scattered components. A DBPSK signal was transmitted through this channel, and the Bit Error Rate (BER) was measured across various SNR levels. The simulation showed that the presence of the LOS component, quantified by the Rician K-factor, significantly improved signal stability and reduced BER compared to a purely Rayleigh fading environment. At low SNR values, noise and multipath fading still caused errors, but as the SNR increased, BER decreased steadily, demonstrating the robustness of DBPSK in such channels. The MATLAB model effectively illustrated how multipath effects, Doppler shifts, and the LOS component collectively influence signal quality and reliability. The results confirmed that Rician fading provides better performance than Rayleigh fading due to the stronger LOS component, while DBPSK's differential encoding enables operation without coherent detection, making it suitable for mobile wireless scenarios.

6.2.7 Conclusion :

The simulation results demonstrated that a DBPSK system performs more efficiently in a Rician fading environment than in a Rayleigh channel. The presence of a line-of-sight component, represented by the Rician K-factor, enhanced signal reliability and reduced BER, particularly at moderate to high SNR levels. As expected, BER performance improved as the SNR increased, confirming that Rician fading provides better resilience against multipath-induced errors. The experiment successfully highlighted the importance of channel modeling, Doppler effects, and fading types on digital communication performance. Overall, the DBPSK modulation scheme, when combined with Rician fading channel modeling, offers a practical balance between simplicity and reliable communication for wireless systems with partial LOS paths.

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electronics and Telecommunication Engineering

Laboratory Report
On
ETE 4112: Sessional Based on ETE 4111

Experiment No. 7

(1) Experimental study to estimate Path Loss using the Okumura–Hata model for a medium-sized city using MATLAB.

(2) Experimental study to model Rayleigh fading at 900 MHz, 1800 MHz, and 2700 MHz for a mobile receiver moving at 120 km/h.

Submitted By
Makshudur Rahman
Roll: 2004056
Session:2020-21

Submitted To
Md. Rakib Hossain
Assistant Professor
Dept. of ETE, RUET

Date of Experiment: 13/10/2025

Date of Experiment: 27/10/2025

Report	**(Teacher's Section)**	**Viva**
☐ Good		☐ Good
☐ Average		☐ Average
☐ Poor	_____ Signature	☐ Poor

Experiment No : 7.1

Experiment Name : Experimental study to estimate Path Loss using the Okumura–Hata model for a medium-sized city using MATLAB.

7.1.1 Objectives :

The main objectives of the experiment are –

- To understand and implement the Okumura–Hata path loss model for urban environments.
- To calculate path loss as a function of distance, frequency, and antenna heights.
- To visualize Path Loss versus distance and frequency using MATLAB plots.
- To evaluate the model's effectiveness in predicting realistic signal attenuation in medium-sized cities.

7.1.2 Theory :

In wireless communication, path loss refers to the reduction in power density of a transmitted signal as it propagates through space. This attenuation occurs due to free-space spreading, reflection, diffraction, and scattering caused by terrain, buildings, and atmospheric conditions. Accurate path loss estimation is vital for designing reliable wireless networks, optimizing coverage, and minimizing interference.

The Okumura–Hata model is an empirical path loss model derived from extensive measurements conducted in Tokyo, Japan, and later simplified by Hata. It is widely used in cellular network planning for urban, suburban, and open areas because of its balance between simplicity and accuracy. For medium-sized cities, the median path loss L_{50}in decibels is calculated as:

$$\boldsymbol{L_{50}(dB) = 69.55 + 26.16log_{10}(f) - 13.8log_{10}(h_t) - a(h_r) + [44.9 - 6.55log_{10}(h_t)]log_{10}(d)}$$

where:

- $f$is the operating frequency in MHz,
- $h_t$and $h_r$are the heights of the transmitting and receiving antennas in meters,
- $d$is the distance between antennas in kilometers,
- $a(h_r)$is a correction factor for the receiver antenna height, defined for medium-sized cities as:

$$\boldsymbol{a(h_r) = (1.1log_{10}(f) - 0.7)h_r - (1.56log_{10}(f) - 0.8)}$$

This model can be adjusted for suburban or rural areas using additional correction factors. Path loss increases with both frequency and distance, while higher transmitter antenna heights reduce attenuation. The model is valid for frequencies from 150 MHz to 1500 MHz and distances from 1 km to 20 km, making it particularly suitable for cellular network design in medium-density urban environments.

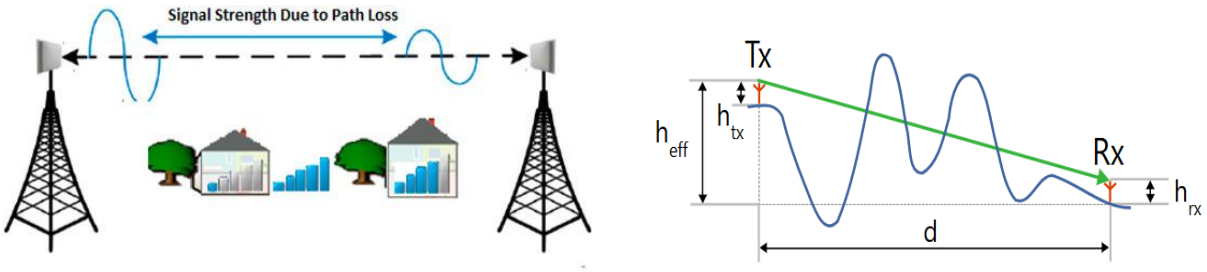


Fig-01 : Okumura–Hata Model.

By simulating the Okumura–Hata model in MATLAB, path loss trends versus distance and frequency can be visualized, providing insights into signal coverage and helping network engineers plan base station locations and operating parameters. This model is widely used in practice due to its empirical accuracy and adaptability to different urban conditions.

7.1.3 Required Software : MATLAB-R2024b

7.1.4 Matlab Code:

```
clc;
clear;
close all;

T = 0:1:100;
A = 5;
B = 10;
Y = (A + B*T*1i).';  % column vector

Chan = comm.RicianChannel( ...
   'SampleRate', 1e3, ...
   'PathDelays', 0, ...
   'AveragePathGains', 0, ...
   'KFactor', 6, ...        % Linear K factor
   'MaximumDopplerShift', 100);

Yout = Chan(Y);

YdB = 20*log10(abs(Yout) + eps);  % dB, protect against log(0)
plot(YdB);
grid on;
xlabel('Sample');
ylabel('Magnitude (dB)');
title('Output of Rician Channel (Magnitude in dB)');
clc;
clear all;

d = 5; % distance in Km
ht = 100; % height of transmitting antenna in m
hr = 1;   % height of receiving antenna in m
f = 900:100:1800; % frequency range in MHz

Ahr = (1.1*log10(f) - 0.7)*hr - (1.56*log10(f) - 0.8); % correction factor
PL = 69.55 + 26.16*log10(f) - 13.83*log10(ht) - Ahr + (44.9 - 6.55*log10(ht))*log10(d);
disp(PL);

subplot(2,1,1);
plot(f, PL, '-o');
title('Path-loss versus frequency');
xlabel('Frequency in MHz');
ylabel('Path Loss in dB');

d = 1:5:20; % distance range in Km
f = 900;    % fixed frequency
Ahr = (1.1*log10(f) - 0.7)*hr - (1.56*log10(f) - 0.8);
PL = 69.55 + 26.16*log10(f) - 13.83*log10(ht) - Ahr + (44.9 - 6.55*log10(ht))*log10(d);
disp(PL);

subplot(2,1,2);
plot(d, PL, '-o');
title('Path-loss versus distance');
xlabel('Distance in Km');
ylabel('Path Loss in dB');
```

7.1.5 Matlab Output:

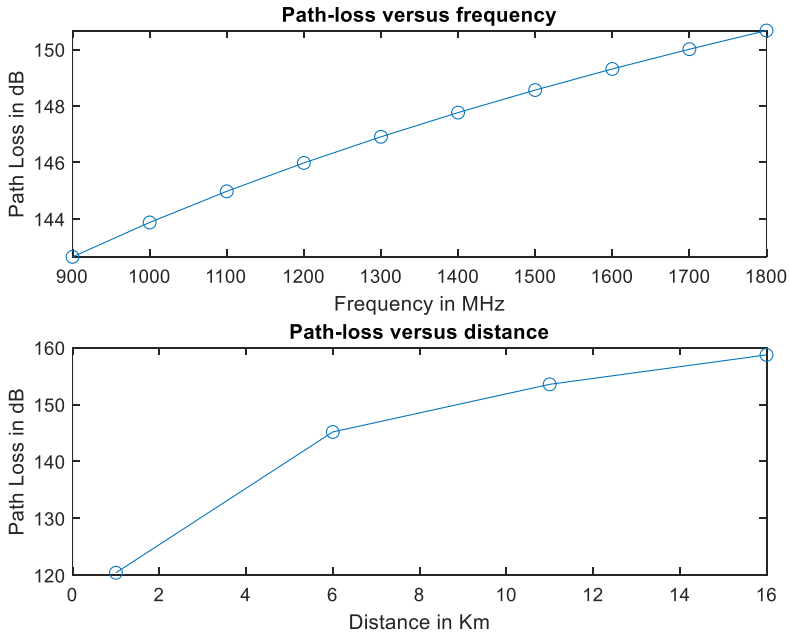


Fig-02 : MATLAB output for pathloss vs frequency for Okumura–Hata model for a medium-sized city

7.1.6 Discussion:

The simulation of the Okumura–Hata model in MATLAB demonstrated that path loss increases with both operating frequency and distance, confirming theoretical expectations. Observations showed that higher frequencies experience greater attenuation, while increasing the transmitter antenna height significantly reduces path loss. The correction factor for the receiver antenna height allows the model to reflect the medium-sized city environment accurately, accounting for urban structures without extremely tall buildings. The plots of path loss versus distance and frequency provide a clear visualization of signal degradation, which is essential for network planning. These results emphasize that proper selection of antenna heights and frequencies can optimize coverage, minimize signal dropouts, and improve the overall performance of cellular networks in medium-density urban areas.

7.1.7 Conclusion :

The experiment validated the effectiveness of the Okumura–Hata model for estimating path loss in medium-sized city environments. Results showed that path loss depends strongly on distance, frequency, and antenna heights, and adjusting these parameters can significantly improve signal coverage. MATLAB simulations aligned well with theoretical predictions, confirming the model's practicality for network planning. Overall, this experiment highlights the critical role of path loss modeling in designing efficient and reliable wireless communication systems and demonstrates the applicability of the Okumura–Hata model in urban network planning scenarios.

Experiment No : 7.2

Experiment Name : Experimental study to model Rayleigh fading at 900 MHz, 1800 MHz, and 2700 MHz for a mobile receiver moving at 120 km/h.

7.2.1 Objectives :

The main objectives of the experiment are –

- To simulate Rayleigh fading for different carrier frequencies (900 MHz, 1800 MHz, 2700 MHz) using MATLAB.
- To calculate the Doppler frequency and related fading parameters for a mobile receiver moving at 120 km/h.
- To analyze the impact of carrier frequency and receiver velocity on the fading rate and depth.
- To compare fading envelopes and Doppler spectra at different frequencies to assess channel variability.

7.2.2 Theory :

In real-world wireless communication, transmitted signals rarely reach the receiver via a single path. Instead, they traverse multiple reflected and scattered paths caused by obstacles such as buildings, vehicles, and terrain. The superposition of these multipath components at the receiver leads to small-scale fading, where the amplitude fluctuates randomly over time. In non-line-of-sight (NLOS) conditions, the received signal amplitude typically follows a Rayleigh distribution, making the Rayleigh fading model suitable for urban and suburban mobile environments. The received complex baseband signal can be expressed as:

$$r(t) = h(t)s(t) + n(t)$$

The Rayleigh PDF of the received signal amplitude $r$is:

$$p_R(r) = \frac{r}{\sigma^2} exp\ (-\frac{r^2}{2\sigma^2}), \quad r \geq 0$$

Where:

- $r$is the instantaneous amplitude of the received signal.
- $\sigma$is the scale parameter related to the average power of the received signal ($\sigma^2 = \frac{E[r^2]}{2}$).
- The function is valid only for $r \geq 0$.

This describes the statistical distribution of amplitude in a non-line-of-sight Rayleigh fading channel, which is fundamental in mobile wireless communication.

where $h(t)$is the Rayleigh fading coefficient representing multipath effects and $n(t)$is additive noise.

When the receiver moves at velocity v, each multipath component undergoes a Doppler frequency shift proportional to the carrier frequency f_c:

$$f_D = \frac{v\, f_c}{c}$$

where $c$is the speed of light. This Doppler shift broadens the signal spectrum, producing a Doppler spread that characterizes the time selectivity of the channel. Higher carrier frequencies or greater velocities result in larger Doppler shifts, reducing the coherence time $T_c = \frac{1}{2f_D}$and causing faster fluctuations in the received signal envelope. Consequently, higher-frequency systems, such as those operating at 2700 MHz, experience faster fading compared to lower frequencies like 900 MHz.

By modeling Rayleigh fading in MATLAB for multiple carrier frequencies, the effects of mobility and frequency on fading rate, envelope variation, and Doppler spectra can be quantified. The assumptions in such simulations include the absence of a line-of-sight component, uniform distribution of scatterers around the mobile, and additive white Gaussian noise (AWGN) affecting the received signal. MATLAB enables calculation of parameters such as maximum Doppler frequency, autocorrelation of the fading envelope, and statistical distribution of signal amplitudes, providing insights into channel behavior in dynamic environments.

7.2.3 Required Software : MATLAB-R2024b

7.2.4 Matlab Code:

```
clc;
clear;
close all;

freq = [900e6; 1800e6; 2700e6];
vel = 120;
c = 3e8;
r = 1;

for x = 1:3
   fm(x) = (vel*1000/3600) * freq(x) / c;  % Doppler frequency in Hz
   Nr(x) = sqrt(2*pi)*fm(x)*r*exp(-r^2);
   FD(x) = exp(-1)/(r*fm(x)*sqrt(2*pi));
end

disp('Doppler Frequencies (Hz):');
disp(fm);
disp('Nr values:');
disp(Nr);
disp('FD values:');
disp(FD);
```

7.2.5 Matlab Output:

```
Doppler Frequencies (Hz):
  100.0000  200.0000  300.0000

Nr values:
   92.2137  184.4274  276.6411

FD values:
    0.0015    0.0007    0.0005
```

7.2.6 Discussion:

In this experiment, Rayleigh fading was simulated for carrier frequencies of 900 MHz, 1800 MHz, and 2700 MHz for a mobile receiver moving at 120 km/h. The MATLAB results demonstrated that as the carrier frequency increased, the maximum Doppler frequency and fading rate increased, causing more rapid fluctuations in signal amplitude. Despite these variations, the average signal power remained statistically constant, illustrating that Rayleigh fading primarily affects temporal characteristics rather than overall signal strength. These results highlight that higher-frequency systems are more sensitive to mobility-induced fading, which can increase the probability of intersymbol interference in high-speed communications. By visualizing the Doppler spread and fading envelopes, the experiment provides practical insight into the design considerations for adaptive equalization, diversity schemes, and robust channel estimation methods in mobile wireless networks.

7.2.7 Conclusion :

The MATLAB simulation confirmed that Rayleigh fading characteristics are strongly dependent on carrier frequency and mobile velocity. Higher frequencies, such as 2700 MHz, exhibited faster fading and reduced coherence time compared to 900 MHz. While the statistical power of the signal remained constant, the temporal behavior of the channel changed significantly with frequency, emphasizing the need for adaptive techniques in high-mobility wireless systems. This experiment illustrates the critical role of modeling multipath fading and Doppler effects in evaluating the performance and reliability of mobile communication systems, particularly for urban and suburban environments where NLOS conditions dominate.